Monopolistically Skewed Business Cycles: On the Origins of Distributional Fluctuations*

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Abstract

A large literature offers competing, often piecemeal explanations for business cycles, invoking diverse shocks and transmission mechanisms. To provide a unified account, we develop an equilibrium model of monopolistic competition in which a single aggregate cost shock jointly drives mean growth, dispersion, and skewness. Heterogeneous exposure spreads the shock across firms, while demand curvature makes their responses asymmetric. We link distributional business-cycle transmission to demand theory and derive these responses under general conditions on demand shape. The model reproduces established business-cycle facts and two opposing size gradients that we document among U.S. public firms: larger firms have lower, more stable growth dispersion but larger cyclical fluctuations in skewness. We estimate that responsiveness roughly halves and normalized curvature magnitude more than doubles from the smallest to the largest size decile, and we confirm the theory's predictions using identified aggregate shocks and a COVID event study.

Keywords: business cycle, skewness, demand curvature, shock propagation
JEL Codes: E32, D21, L11

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1 Introduction

Shocks, which come in various shapes and forms, drive the business cycle: adverse TFP shocks "pluck" the economy below its expansion path (Dupraz et al., 2025).[1] Uncertainty shocks make firms postpone hiring and investment (Bloom, 2009).[2] Downside-risk shocks cause deep and prolonged downturns (Salgado et al., 2026). The nature and origin of these shocks determine the severity and persistence of recessions, with far-reaching implications for welfare and stabilization policy.

Accordingly, alongside a growing list of shock types, a large literature has proposed a smorgasbord of mechanisms and cyclical patterns in the growth distribution. These explanations often assign distinct drivers to different patterns: uncertainty shocks for rising dispersion, downside-risk shocks for left skewness. But is this multiplicity really required? Could a *single aggregate cost shock*, transmitted through firms' equilibrium responses to the shock ("policies"), account for how the distribution of firm growth moves over the cycle, thereby unifying much of the existing evidence? We find that it can. We establish this result in a general class of monopolistic competition models, without resorting to adjustment costs or financial or labor-market frictions.

Our theory reconciles a broad set of empirical observations about cross-sectional growth: recessions bring lower mean growth and more dispersed, left-skewed growth distributions (Bloom et al., 2018; Salgado et al., 2026).[3] Uncertainty shocks coincide with adverse productivity shocks (Bloom et al., 2018). Firms respond more strongly to adverse than to favorable shocks (Ilut et al., 2018). Large firms react less to idiosyncratic cost shocks, yet their average growth rates move with the cycle much like those of smaller firms (Amiti et al., 2019; Crouzet and Mehrotra, 2020). Larger firms also have lower and more stable growth dispersion, but far larger swings in skewness (this paper). Cyclical dispersion and skewness therefore need not reflect separate shocks: they arise from how price-setting firms transmit a common one, and so carry information about demand curvature and market power.

The mechanism in our model is straightforward. An adverse aggregate shock raises firms' costs on average. Because firms are heterogeneously exposed to the shock, it also widens the distribution of cost changes. Under monopolistic competition, demand curvature makes firms respond concavely to their own cost

[1] The plucking account goes back to Friedman (1993). For related evidence on business-cycle asymmetry, see Sichel (1993), McKay and Reis (2008), and Morley and Piger (2012).

[2] Related mechanisms operate through financial frictions (Arellano et al., 2019) and labor-market frictions (Schaal, 2017).

[3] Left, or negative, skewness describes an imbalance toward large growth shortfalls. Intuitively, the cross-sectional distribution extends farther or carries more weight in its left tail.

shocks: they contract more after adverse shocks than they expand after equally sized favorable shocks. Therefore, mean growth falls, dispersion rises, and left-skewness emerges. The same demand structure also explains heterogeneity across firm sizes. Larger firms respond less strongly but more asymmetrically to the component of the shock transmitted through their firm-specific exposure. Together, weaker responsiveness and greater asymmetry generate smaller dispersion fluctuations but larger swings in skewness, whereas mean growth responds similarly across firm sizes.

We examine these predictions in Compustat data on U.S. public firms, where "small" and "large" refer to relative size within this public-firm sample. We first establish a set of stylized facts that connects the observations in previous work to our sample. Mean growth falls sharply in recessions and rebounds during recoveries, while dispersion rises as growth moves away from trend. Skewness is procyclical, and its fluctuations increase with firm size even as dispersion and its fluctuations decline. We call this new finding the *opposing size gradients in growth moments* (*size gradients*, for short) and use it to discipline the theory.

We then test the model's predictions using identified aggregate shocks, a COVID event study, and estimates of firms' equilibrium policy functions. We find that firms' growth responses to six aggregate shocks are generally more dispersed among small firms and more negatively skewed among large firms. In the COVID event study, greater exposure dispersion accompanies a larger increase in growth variance and a sharper decline in skewness, holding industry–size composition fixed. Joint estimates of policy slopes, normalized curvature (curvature divided by the absolute policy slope), and idiosyncratic risk confirm the predicted differences across firm sizes. From the smallest to the largest size decile, responsiveness roughly halves and normalized curvature magnitude more than doubles; the volatility of idiosyncratic cost shocks falls by about a third. These differences persist under economy-wide size rankings and within-industry comparisons.

The model features monopolistic competition with general homothetic demand and isoelastic production costs. Productivity differences determine firms' baseline sizes, placing them at different operating quantities on a common inverse-demand curve. Changes in an aggregate cost factor generate different effective shocks across firms through heterogeneous exposure, following the evidence in Davis et al. (2025). For example, a rise in diesel prices affects a shipping business that relies on diesel trucks more strongly than one that uses electrified railways.

Firms' optimal responses to their effective cost shocks depend on how demand elasticity changes with output. Marshall's Second Law of Demand (MSLD) states

that demand becomes less price-elastic as a firm expands along its demand curve. Marshall's Strong Second Law of Demand (MSLD*) additionally requires that marginal revenue becomes more elastic in absolute value as output expands. Under it, profit maximization yields concave log-output and log-real-revenue policies: cost cuts induce smaller expansions than the contractions caused by equally sized cost increases. In contrast, a price taker producing with the same technology has a linear log-output policy with a steeper slope. Price setting therefore both reduces responsiveness and, under MSLD*, makes adjustment asymmetric. We derive from firms' pricing incentives the concavity that Ilut et al. (2018) use to explain asymmetric growth and cyclical volatility.

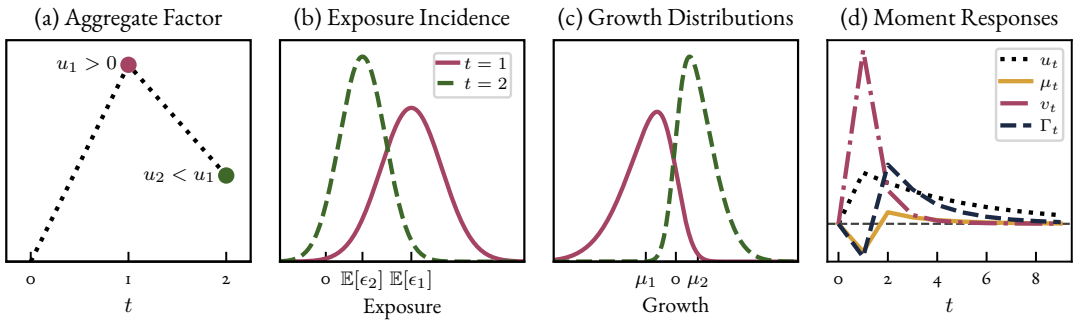
This comparison extends to differences in operating size among price-setting firms. Near a finite choke price (the price at which demand vanishes), prices approach marginal cost and policies locally approach those of the price taker; within this region, lower-cost firms operate at larger quantities and have flatter policies but greater normalized curvature magnitude. The common demand structure thus makes larger firms less responsive but more asymmetric.

These policy differences govern responses to deviations from cross-sectional average exposure. The common component, equal to cross-sectional average exposure multiplied by the aggregate shock, induces the same proportional adjustment across firms. Homothetic demand separates this common adjustment from firms' nonlinear policies, giving size groups the same first-order change in mean growth. The equilibrium therefore reconciles weaker responses to idiosyncratic cost shocks among larger firms (Amiti et al., 2019) with similar mean cyclicality across firm sizes (Crouzet and Mehrotra, 2020).

Figure 1 illustrates how these policies translate an aggregate cost shock into joint changes in mean growth, dispersion, and skewness. Consider firms with the same baseline size and policy but different exposures. Figure 1(a) shows an aggregate cost increase followed by a partial reversal. Heterogeneous exposure spreads the cost increase across firms, as Figure 1(b) illustrates. These firm-specific cost changes pass through concave policy functions, producing left-skewed growth on impact in Figure 1(c) (solid curve, $t = 1$). As the shock recedes, firms that contracted most recover most, producing right-skewed growth. Figure 1(d) traces the resulting moment dynamics: dispersion rises during both sharp contractions and sharp recoveries, while skewness switches from negative to positive.

Our model formalizes this intuition through relationships between policy coefficients and growth moments. The absolute policy slope measures responsiveness;

Figure 1: Distributional Shifts in Theory



Note: Panel (a) shows an aggregate cost increase followed by a partial reversal. Panel (b) illustrates how heterogeneous exposure distributes the resulting cost shocks across firms, with $\epsilon_{it} = \lambda_i u_t$, abstracting from idiosyncratic shocks. Panel (c) shows the corresponding growth distributions: left-skewed on impact and right-skewed during recovery. Panel (d) traces the aggregate factor u_t, mean growth μ_t, variance v_t, and skewness Γ_t as the shock dissipates.

normalized curvature measures response asymmetry. Growth variance is approximately proportional to the squared policy slope and growth skewness to normalized curvature. Flatter policies therefore imply lower and less variable growth variance, while greater normalized curvature magnitude produces larger skewness fluctuations. Because homotheticity separates the common growth component from the nonlinear policies, we can characterize these moment responses using coefficients evaluated at the no-shock equilibrium. By contrast, this separation is not possible with the location channel in Straub and Ulbricht (2019) and Ilut et al. (2018), where common movements shift firms along nonlinear response functions and change their local slopes and curvatures.

We thus link the literature on demand shape (Kimball, 1995; Weyl and Fabinger, 2013; Mrázová and Neary, 2017; Melitz, 2018) to distributional business-cycle transmission. In this parsimonious equilibrium framework with general homothetic demand,[4] we derive conditions on demand curvature under which a single aggregate cost shock reproduces the stylized facts. We discuss competing explanations based on shocks and technology in detail, explaining why demand provides the most plausible economic foundation for these patterns.[5]

Building on this framework, we contribute to the literature on identifying shocks and their transmission (Jurado et al., 2015; Berger and Vavra, 2019; Ludvigson

[4]For flexible homothetic demand systems beyond CES, see Feenstra (2003) and Matsuyama and Ushchev (2017). For a general treatment of monopolistic competition, see Parenti et al. (2017).

[5]Production networks provide another source of asymmetric transmission; see Miranda-Pinto et al. (2023), Dew-Becker and Vedolin (2025), and Kamepalli et al. (2025).

et al., 2021). Under common exposure restrictions, we use the joint dynamics of growth variance and skewness to distinguish differences in responsiveness and asymmetry from differences in idiosyncratic cost volatility across the size distribution. Accounting for idiosyncratic risk is essential because it mechanically raises dispersion and dampens standardized skewness.

We also contribute to the literature on heterogeneous shock transmission (Ilut et al., 2018; Berger and Vavra, 2019; Davis et al., 2025). We show that concave responses differ systematically in slope and curvature across firm sizes and that heterogeneous exposure drives differences in both dispersion and skewness. Together, these findings explain how a single aggregate shock generates different growth distributions across size groups.

Finally, we connect our findings to the literature on variable markups and misallocation (Dhingra and Morrow, 2019; Edmond et al., 2023). In our framework, the demand properties that make larger firms less responsive but more asymmetric also generate markup differences that distort production. Our growth-moment evidence provides a new way to discipline these properties.

Section 2 documents the growth-distribution facts. Section 3 develops the model and derives the policy shapes implied by demand. Section 4 derives growth moments, interprets the stylized facts, and examines alternative mechanisms. Section 5 recovers policy differences and tests the predictions using identified shocks and measured exposures. Section 6 concludes.

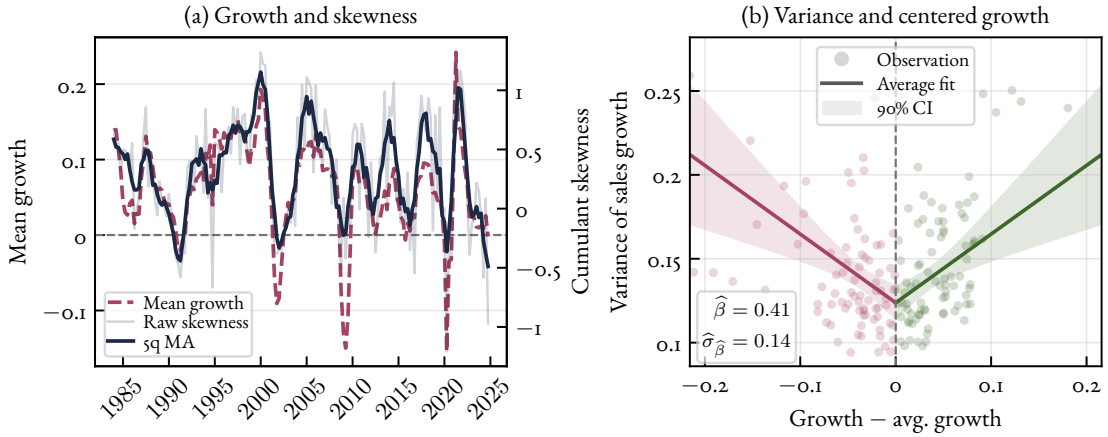
2 Stylized Facts

We begin with three facts about the cross section of firm growth. Mean growth falls sharply in recessions; dispersion is low near trend, rises as aggregate growth deviates from trend, and is generally higher for small firms. Skewness is procyclical, with larger fluctuations among large firms.[6]

Data and moments Our baseline sample covers U.S. public firms in Compustat from 1984Q1 through 2024Q4, with the top and bottom 1 percent of firm growth observations trimmed each quarter to limit the influence of outliers. Appendix C

[6]Early studies of cyclical firm-growth distributions include Higson et al. (2002, 2004). For evidence and theories concerning scale dependence and the shape of firm-growth distributions, see Sutton (1997); Bottazzi and Secchi (2006); Sterk et al. (2021).

Figure 2: Growth, Cumulant Skewness, and Dispersion Around Trend



Note: Panel (a) plots mean year-on-year real-sales growth, cumulant skewness, and its centered five-quarter moving average. Panel (b) plots variance against mean growth relative to its time average. The lines show a symmetric V-shaped fit. Shaded regions are 90% Newey–West confidence intervals with four lags.

provides further details on sample construction.[7] To study how growth patterns vary with firm size, we measure size throughout the main text by average real quarterly sales over $t-4,\ldots,t-1$ and compare firms within two-digit NAICS industries. In this section, these comparisons use size deciles formed within each industry every quarter.

Let $R_{it} = p_{it}q_{it}$ be firm i's nominal sales at time t, P_t the aggregate price level, and $r_{it} = R_{it}/P_t$ real sales. We measure firm growth by the year-on-year change $g_{it} = \log r_{it} - \log r_{i,t-4}$. To track the mean, dispersion, and skewness of firm growth, we compute in each quarter the cross-sectional mean $\bar{g}_t$, variance v_t, and cumulant skewness

$$\Gamma_t := \frac{\mathbb{E}_i[(g_{it} - \bar{g}_t)^3]}{v_t^{3/2}}. \tag{1}$$

Negative skewness indicates a longer or heavier left tail, and positive skewness the reverse.

Stylized Fact 1: ***Mean growth follows a 'plucking' pattern, normally fluctuating around an expansion trend and occasionally falling sharply around recessions.*** The dashed red line in panel (a) of Figure 2 shows that mean firm sales growth usually stays near its expansion trend but falls sharply

[7]Table C.1 reports summary statistics and compares these firms with the Quarterly Financial Report (QFR) manufacturing sample of Crouzet and Mehrotra (2020). The comparison makes clear that our sample captures the upper part of the firm-size distribution, within which we study size heterogeneity.

around recessions. These declines are often followed by sharp rebounds, most visibly after the 2009 and 2020 contractions.[8] This pattern echoes the familiar asymmetry of aggregate fluctuations: contractions tend to be sharp and deep, while expansions are gradual (Sichel, 1993; Morley and Piger, 2012; Jensen et al., 2020). It also resembles the plucking dynamics of Dupraz et al. (2025), in which the economy is pulled below trend and subsequently recovers.

Stylized Fact 2: ***Growth dispersion rises when mean growth moves away from trend. The average level and the time-series variation of dispersion both decline with firm size.*** Panel (b) of Figure 2 plots the cross-sectional variance of firm growth against deviations of mean growth from its time average. Dispersion is lowest near average growth and rises on either side, tracing a V-shape. We summarize this pattern with $v_t = \alpha + \beta|\bar{g}_t - \bar{g}| + \varepsilon_t$, where $\bar{g}$ is the time average of mean growth. The estimated coefficient is positive and significant. High dispersion is thus not confined to recessions: sharp contractions and sharp rebounds both come with greater differences in firm growth.

Dispersion also exhibits a strong size gradient, both in its average level and in its variation over time. To show this, we compare these moments across the within-industry size deciles.[9] Panel (a) of Figure 3 shows that both the time-series mean $\mathbb{E}_{\text{time}}[v_{jt}]$ and standard deviation $\text{SD}_{\text{time}}(v_{jt})$ decline monotonically with size, where v_{jt} is the cross-sectional variance of growth in decile j. Larger firms thus have lower dispersion on average, consistent with Crouzet and Mehrotra (2020), and their dispersion varies less over time. So, the size gradient in levels extends to fluctuations in dispersion itself.

Stylized Fact 3: ***Cumulant skewness is strongly procyclical, and its time-series variability rises steeply with firm size.*** Firm-growth skewness is strongly procyclical, as emphasized by Salgado et al. (2026).[10] Panel (a) of Figure 2 confirms this pattern for cumulant skewness: the correlation between $\bar{g}_t$ and Γ_t is 0.78. Downturns come with a more left-skewed distribution of firm growth; recoveries with a more right-skewed one.

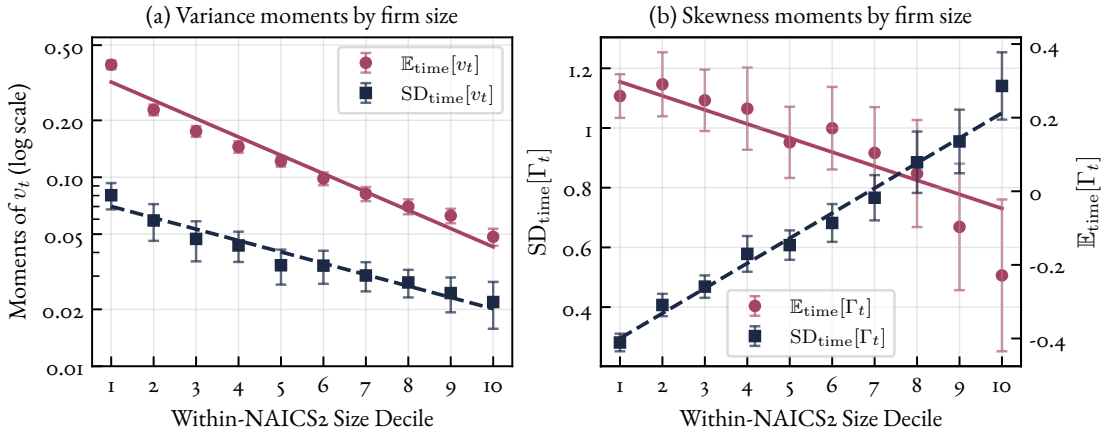
The same comovement appears in every size group, with correlations between

[8] Appendix Figure D.1 shows the mean-growth distribution and the contraction–recovery chronology.

[9] As mentioned, the within-industry size ranks are our baseline in the main analysis. Appendix Figure D.2 shows that the opposing size gradients are nearly unchanged when firms are instead ranked in the economy-wide size distribution, without using industry information.

[10] Related evidence concerns forward-looking financial skewness (Dew-Becker, 2024) and expected aggregate macroeconomic skewness (Iseringhausen et al., 2026).

Figure 3: Opposing Size Gradients in Growth Moments



Note: Circles and squares report time-series means and standard deviations of growth variance in Panel (a) and cumulant skewness in Panel (b). Bars are 90% delta-method Newey–West intervals with four lags. Lines show log-linear fits in panel (a) and linear fits in panel (b), weighted by the inverse variance of the estimated moments.

each decile's skewness and aggregate mean growth ranging from 0.46 to 0.71. What changes with size is the amplitude of skewness fluctuations. Panel (b) of Figure 3 reports the time-series mean and standard deviation of Γ_{jt} in each size decile j. The standard deviation rises monotonically from about 0.28 in the smallest decile to 1.14 in the largest. Average skewness tends to decline with size, though this pattern is less precise.[11] The key finding is thus the opposing size gradients: larger firms have lower and more stable dispersion, yet larger fluctuations in skewness.[12]

We next ask whether heterogeneous exposure and demand curvature can jointly explain these patterns.

3 Model

We begin with an industry equilibrium in which a common cost shock reaches firms with different intensities. The equilibrium implies firm policies that separate a common scale change from firms' responses to heterogeneous incidence. We then derive how demand shapes these policies and how their responses vary with firm size. Section 4 links these policy properties to the stylized facts. Because our data

[11] We note a mildly positive average skewness over time. This may reflect secular innovation, entry and exit, or skewed idiosyncratic shocks. For innovation-based sources of right-skewed growth, see, for example, the literature on quality-ladder models or, more recently, Berlingieri et al. (2025). The relevant fact for our mechanism is the cyclical movement of skewness with aggregate growth.

[12] The broad patterns also appear in the Compustat Global sample of non-U.S. firms (Appendix Figures D.3 and D.4).

measure real sales rather than physical output, we formulate the model for both output q_i and real sales r_i.

3.1 Industry Model Setup

Firms compete in a monopolistically competitive market à la Dixit and Stiglitz (1977), with homothetic demand.[13] They differ in their baseline productivity, which determines their size, and in their exposure to an aggregate cost factor.

Households. A representative consumer spends a fixed nominal amount $E > 0$ on goods $\boldsymbol{q} = (q_i)_{i \in I}$ at prices $\boldsymbol{p}$, where I is a unit-measure continuum of varieties. Boldface denotes vectors across varieties. Preferences are $U(\boldsymbol{q}) = F(h(\boldsymbol{q}))$, where F is strictly increasing and h is strictly increasing, concave, and homogeneous of degree one.[14]

Homothetic preferences imply the following demand properties (Parenti et al., 2017). The expenditure function is $e(\boldsymbol{p}, \bar{h}) = P(\boldsymbol{p})\bar{h}$, with $P(\boldsymbol{p}) = e(\boldsymbol{p}, 1)$, where the ideal price index P is homogeneous of degree one. Demand for variety i is

$$q_i(\boldsymbol{p}, E) = \frac{E}{P(\boldsymbol{p})} D_i\left(\frac{\boldsymbol{p}}{P(\boldsymbol{p})}\right), \qquad D_i(\boldsymbol{z}) = \frac{\partial P(\boldsymbol{z})}{\partial z_i}.$$

The corresponding inverse demand is

$$p_i(\boldsymbol{q}, E) = E\frac{h_i(\boldsymbol{q})}{h(\boldsymbol{q})}, \qquad h_i(\boldsymbol{q}) = \frac{\partial h(\boldsymbol{q})}{\partial q_i}.$$

Euler's theorem gives $\int_I p_i(\boldsymbol{q}, E) q_i \, di = \frac{E}{h(\boldsymbol{q})} \int_I h_i(\boldsymbol{q}) q_i \, di = E$. Concavity of h makes $\boldsymbol{q}$ utility-maximizing at these prices and expenditure, so inverse demand gives the prices that clear all product markets. The associated price index and relative prices satisfy $P(\boldsymbol{p}(\boldsymbol{q}, E)) = E/h(\boldsymbol{q})$ and $p_i(\boldsymbol{q}, E)/P(\boldsymbol{p}(\boldsymbol{q}, E)) = h_i(\boldsymbol{q})$.

Technology. There is a unit mass of firms, each producing one variety. Production costs are isoelastic in output, with exponent $\eta > 1$. Firm i has a baseline cost shifter $\phi_i > 0$, interpreted as inverse productivity. The aggregate cost factor is $u \geq 0$, and firm i's fixed exposure to this factor is $\widetilde{\lambda}_i$. Write $\bar{\lambda} = \mathbb{E}_i[\widetilde{\lambda}_i] \geq 0, \quad \lambda_i = \widetilde{\lambda}_i - \bar{\lambda}$. Changes in u alter aggregate conditions; differences in λ_i determine how strongly they affect each firm. Centered exposures have a symmetric density Λ, bounded

[13] For flexible homothetic demand systems beyond CES, see Kimball (1995); Feenstra (2003); Matsuyama and Ushchev (2017).

[14] On a neighborhood of the counterfactual no-shock output vector defined below, $h \in C^5$, its derivatives through fifth order are uniformly bounded across varieties, and $h_i > 0$.

support, and variance $\sigma_\lambda^2 > 0$. For example, firms may differ in their exposure to an input-price shock through their production networks. The measured exposures of Davis et al. (2025) provide empirical motivation for such heterogeneity.

Each firm also receives a purely idiosyncratic shock κ_i, independent of λ_i and independently and identically distributed across firms. Its distribution is symmetric, with bounded support, mean zero, and variance σ_κ^2. The total cost shock is

$$\widetilde{\epsilon}_i = \underbrace{\kappa_i + \lambda_i u}_{\epsilon_i \text{ (centered component)}} + \underbrace{\bar{\lambda} u}_{\text{common component}}, \qquad \bar{\lambda}, u \geq 0. \tag{2}$$

The centered component has standard deviation $\sigma_\epsilon = (\sigma_\kappa^2 + u^2 \sigma_\lambda^2)^{1/2}$. Average exposure determines the common cost movement, while exposure heterogeneity determines the spread around it.

Firm i's production cost is $C_i(q_i; u, \epsilon_i, \phi_i) = \phi_i e^{\bar{\lambda} u + \epsilon_i} q_i^\eta, \quad \eta > 1$. Because u and $\bar{\lambda}$ are nonnegative, the common component raises costs relative to the absence of aggregate shocks. As this component unwinds, activity recovers. This gives the aggregate factor the plucking interpretation motivated by Stylized Fact 1.

Information structure. Firms observe u and their own κ_i, and know their own exposure λ_i, average exposure $\bar{\lambda}$, and the baseline cost vector $\boldsymbol{\phi}$. They do not observe rivals' exposures or idiosyncratic shocks. Each firm chooses output using its own realized cost, while conjecturing that its rivals face only the common cost component $\bar{\lambda} u$. The equilibrium definition below combines this decision rule with market clearing at realized quantities.

3.2 Equilibrium

3.2.1 Definition

We begin with the firm's problem at the counterfactual no-shock benchmark, where $u = \kappa_i = 0$ and hence $\epsilon_i = \epsilon^\star = 0$ for every firm. Baseline cost shifters ϕ_i can still differ, and firm i's marginal cost is $\eta \phi_i q_i^{\eta-1}$. Firm i's nominal revenue at the full output vector $\boldsymbol{q}$ is

$$R_i(q_i; \boldsymbol{q}_{-i}) \equiv p_i(\boldsymbol{q}, E) q_i = E \frac{q_i h_i(\boldsymbol{q})}{h(\boldsymbol{q})}.$$

Revenue is homogeneous of degree zero in the full output vector. Let $\bar{\boldsymbol{q}} = \bar{\boldsymbol{q}}(\boldsymbol{\phi}) \gg 0$ collect the mutually consistent profit-maximizing outputs in this no-shock counterfactual:

$$\bar{q}_i = \arg\max_{x > 0} \left\{ R_i(x; \bar{\boldsymbol{q}}_{-i}) - \phi_i x^\eta \right\}. \tag{3}$$

We maintain uniqueness and interiority throughout: firms do not optimally shut down production, and there is no entry margin.

To construct firms' conjectures about rivals' output, fix a counterfactual in which every firm has average exposure and no idiosyncratic shocks: $\lambda_i = \kappa_i = 0$. Thus $\epsilon_i = 0$ even as u varies. Homothetic demand and isoelastic costs imply a proportional output adjustment, $q_i(u) = s(u)\bar{q}_i, \quad s(u) = e^{-\bar{\lambda}u/\eta}$. To see this, suppose rivals produce $s(u)\bar{\boldsymbol{q}}_{-i}$ and write the firm's own output as $s(u)x$. Homogeneity gives $R_i(s(u)x; s(u)\bar{\boldsymbol{q}}_{-i}) = R_i(x; \bar{\boldsymbol{q}}_{-i})$, while $e^{\bar{\lambda}u}s(u)^\eta = 1$. The firm's profit-maximization problem therefore reduces to Equation (3).

This counterfactual separates the common output adjustment from the relative adjustments caused by heterogeneous shocks. A uniform cost change scales all firms' quantities by the same proportion, even when their baseline productivities differ. Heterogeneous exposure and idiosyncratic shocks break this proportional adjustment: costs change by different amounts across firms, and their policies determine the resulting relative output responses.

Actual rival outputs generally depart from these counterfactual quantities because exposures and idiosyncratic shocks differ across firms. Under the information structure above, firms cannot condition on these rival shocks. Instead, we specify that each firm conjectures rival output $s(u)\bar{\boldsymbol{q}}_{-i}$ and chooses its own output using its realized cost. We call these *aggregate-scale conjectures*. Firms thus optimize against perceived demand, while transactions take place at the prices that clear markets for actual output. This separation between perceived demand and realized market outcomes follows the approach of Negishi (1961).

Definition 1 (Equilibrium under aggregate-scale conjectures)**.** *Fix the public baseline cost vector $\boldsymbol{\phi}$, the aggregate shock u, and the realized vector of firm-specific shocks $\boldsymbol{\epsilon}$. An equilibrium consists of an output vector $\boldsymbol{q} \gg 0$ and prices $\widetilde{\boldsymbol{p}}$ such that each firm conjectures rival output $s(u)\bar{\boldsymbol{q}}_{-i}$ and chooses*

$$q_i = \arg\max_{x>0} \left\{ R_i(x; s(u)\bar{\boldsymbol{q}}_{-i}) - \phi_i e^{\bar{\lambda}u+\epsilon_i} x^\eta \right\}, \tag{4}$$

and markets clear at $\widetilde{p}_i = p_i(\boldsymbol{q}, E), \quad P = P(\widetilde{\boldsymbol{p}})$. Output choices use conjectured rival quantities; market-clearing prices depend on actual quantities.

In Definition 1, firms use the aggregate shock to construct perceived demand, then choose output given their own realized cost. These demand conjectures are correct when firms face only the common cost component. With heterogeneous exposure or idiosyncratic shocks, conjectured and realized demand generally differ. The concept is closely related to the oblivious-equilibrium approach of Weintraub

et al. (2008), in which firms condition on their own state and an aggregate representation of the industry while ignoring competitors' current individual states. Here, u determines the common scale $s(u)$, which each firm combines with its own shock ϵ_i.

This approach reduces firms' informational requirements (Moll and Ryzhik, 2026). Full information would require firms to observe competitors' shocks; rational expectations would at least require knowledge of their distribution, competitors' strategies, and the full demand system. Here, firm i needs the homothetic structure of demand and its own no-shock demand schedule, which it scales using the aggregate shock.[15] Weintraub et al. (2008) show that, under regularity conditions, oblivious equilibrium closely approximates Markov-perfect equilibrium in a dynamic industry model. Their result does not directly apply to our static environment, but illustrates how restricted conditioning can approximate full-state equilibrium.

3.2.2 Policies

We characterize firm-level policies by setting the aggregate shock to zero in Equation (4), while allowing the firm-specific shock ϵ to vary. Write $x_i(\epsilon)$ for the resulting output policy, with $x_i(0) = \bar{q}_i$. The corresponding log-output policy is $\mathcal{Q}_i(\epsilon) = \log x_i(\epsilon)$. Real revenue is output multiplied by the firm's real price, $h_i = p_i/P$. With the aggregate shock set to zero, firm i conjectures that rivals produce their counterfactual no-shock outputs $\bar{\boldsymbol{q}}_{-i}$. Its perceived real price at output $x_i(\epsilon)$ is therefore $h_i(x_i(\epsilon), \bar{\boldsymbol{q}}_{-i})$, giving the conditional log-real-revenue policy

$$\mathcal{R}_i(\epsilon) = \mathcal{Q}_i(\epsilon) + \log h_i(x_i(\epsilon), \bar{\boldsymbol{q}}_{-i}).$$

Realized real revenue can differ because market-clearing prices depend on rivals' actual quantities.

These policies were constructed with the aggregate shock set to zero and rivals held at their counterfactual no-shock outputs. Proposition 1 shows how the same functions describe outcomes when aggregate and firm-specific shocks are present. Log output receives a common additive adjustment; log real revenue also includes the discrepancy between conjectured and realized relative prices. Equation (5) provides the key link between our model and the stylized facts, allowing us to interpret the observed growth-moment patterns through firms' policies.

[15] The model extends to multiple aggregate shocks. Richer public information about industry conditions can therefore be represented by a larger set of observed aggregate factors, without making firm-specific exposures public.

Proposition 1 (Industry Equilibrium)**.** *Suppose preferences are homothetic and Equation* (4) *has a unique interior solution for each firm. Fix the public baseline cost vector* $\boldsymbol{\phi}$*, suppressing it in the notation. Let* $s(u) = e^{-\bar{\lambda}u/\eta}, \quad \boldsymbol{x}(\boldsymbol{\epsilon}) = (x_i(\epsilon_i))_{i\in I}$*. Then:*

1. *Firm output satisfies* $q_i(u, \epsilon_i) = s(u)x_i(\epsilon_i), \quad x_i(0) = \bar{q}_i$*.*

2. *Market-clearing prices and the price index satisfy*

$$\widetilde{p}_i(u, \boldsymbol{\epsilon}) = s(u)^{-1} \frac{E h_i(\boldsymbol{x}(\boldsymbol{\epsilon}))}{h(\boldsymbol{x}(\boldsymbol{\epsilon}))}, \qquad P(u, \boldsymbol{\epsilon}) = s(u)^{-1} \frac{E}{h(\boldsymbol{x}(\boldsymbol{\epsilon}))}.$$

   Realized prices incorporate the baseline cost distribution and the realized cross-section of firm-specific shocks, although firms do not condition on rivals' shocks when choosing output.

3. *Define realized real revenue as* $r_i(u, \boldsymbol{\epsilon}) = \frac{\widetilde{p}_i(u,\boldsymbol{\epsilon}) q_i(u,\epsilon_i)}{P(u,\boldsymbol{\epsilon})}$ *and the cross-price residual as*

$$\zeta_i(\boldsymbol{\epsilon}) = \log h_i(\boldsymbol{x}(\boldsymbol{\epsilon})) - \log h_i(x_i(\epsilon_i), \bar{\boldsymbol{q}}_{-i}).$$

   Log output and log real revenue satisfy

$$\begin{aligned} \log q_i(u, \epsilon_i) &= -\frac{\bar{\lambda}}{\eta} u + \mathcal{Q}_i(\epsilon_i), \\ \log r_i(u, \boldsymbol{\epsilon}) &= -\frac{\bar{\lambda}}{\eta} u + \mathcal{R}_i(\epsilon_i) + \zeta_i(\boldsymbol{\epsilon}). \end{aligned} \tag{5}$$

Price forecast errors. The cross-price residual is unobserved and can complicate inference about firms' policies from revenue growth. From Proposition 1, it can be written as

$$\zeta_i(\boldsymbol{\epsilon}) = \log\left[\frac{\widetilde{p}_i(u, \boldsymbol{\epsilon})}{P(u, \boldsymbol{\epsilon})\, h_i(x_i(\epsilon_i), \bar{\boldsymbol{q}}_{-i})}\right]. \tag{6}$$

The numerator is the realized price; the denominator is the conjectured price expressed at the same price-index level. The price residual arises ex post and does not enter the firm's decision problem. For inference from revenue data, what matters is how this forecast error varies across firms. Proposition A.1 and Remark A.1 in Online Appendix A give sufficient conditions under which the residual is common within a cross section of firms up to second order, with a third-order heterogeneous component. Cross sections may differ in baseline productivity and demand. These conditions support the revenue-moment analysis in Section 4; the output-policy representation does not require them.

3.3 Demand Shape and Firm Responses

We now ask which properties of demand make the equilibrium policies concave and how their responses vary with firm size. We characterize output first, account for relative-price adjustment in real revenue, and then compare firms along a common inverse-demand curve.

3.3.1 Normalized Firm Problem

The functions $\mathcal{Q}$ and $\mathcal{R}$ are independent of u. We can therefore recover the same policies by setting $u = 0$ in Equation (4) and solving the partial-equilibrium problem $\max_{q\geq 0}\{qp(q) - C(q;\epsilon)\}$. Suppressing the firm index, $p(q) \propto h_i(q;\bar{\boldsymbol{q}}_{-i})$ is its residual inverse demand and $C(q;\epsilon) = \phi e^{\epsilon} q^{\eta}$ is its cost, with $\phi > 0$ and $\eta > 1$. We retain isoelastic costs so that curvature in the log-output policy comes from demand adjustment.

We maintain the regularity conditions in Appendix Assumption 1 throughout. These conditions give a unique interior optimum that equates marginal cost and marginal revenue (Appendix Lemma 1). Writing $\mathcal{E}f(q) := qf'(q)/f(q)$ for elasticity and $R'(q) := p(q) + qp'(q)$ for marginal revenue, the first-order condition is

$$\eta\phi e^{\epsilon} q^{\eta-1} = p(q)\Big(1 + \mathcal{E}p(q)\Big) = R'(q). \tag{7}$$

Taking logs gives

$$\epsilon = \log R'(q) - \log\eta - (\eta-1)\log q - \log\phi =: H(\log q). \tag{8}$$

Thus H maps log output into the cost shock, and its inverse is the decreasing log-output policy $\mathcal{Q}$. Demand shapes this policy through the marginal-revenue term in H.

For a price taker, marginal revenue is the fixed price $\bar{p}$, so the first-order condition becomes $\epsilon = \log\bar{p} - \log\eta - (\eta-1)\log q_{\text{pt}} - \log\phi$. This is affine in log output. Its inverse policy $\mathcal{Q}_{\text{pt}}$ is therefore affine in the shock, providing the benchmark for the demand-induced curvature studied next.

3.3.2 Policy Responsiveness and Curvature

We now use the first-order condition to relate policy slopes and curvature to demand properties. We begin with physical output and then turn to real revenue.

Demand-shape objects. To characterize the marginal-revenue term in H, define the elasticity and superelasticity of inverse demand together with the elasticity of marginal revenue:

$$\varepsilon(q) := \mathcal{E}p(q) < 0, \quad \psi(q) := \mathcal{E}\varepsilon(q) = \frac{q\varepsilon'(q)}{\varepsilon(q)}, \quad m(q) := \mathcal{E}R'(q) = \frac{qR''(q)}{R'(q)}. \tag{9}$$

Here, ψ is the *superelasticity* of inverse demand (Melitz, 2018). Let $q^\star := \exp(\mathcal{Q}(0))$ denote output at the no-shock benchmark. We write $\varepsilon^\star := \varepsilon(q^\star)$, $\psi^\star := \psi(q^\star)$, and $m^\star := m(q^\star)$. The following restrictions link demand properties at this reference quantity to policy slope and curvature.

Definition 2 (Marshallian Second-Law Properties)**.**

i) Marshall's Second Law of Demand *(MSLD) holds at $q^\star$ if the absolute elasticity of inverse demand is locally increasing: $-\varepsilon'(q^\star) > 0$.*

ii) Marshall's Strong Second Law of Demand *(MSLD*) holds at $q^\star$ if MSLD holds and the absolute elasticity of marginal revenue is locally increasing: $-m'(q^\star) > 0$.*

iii) Anti-MSLD *holds at $q^\star$ if the absolute elasticity of inverse demand locally decreases, $-\varepsilon'(q^\star) < 0$.* Anti-MSLD* *holds if, additionally, the absolute elasticity of marginal revenue locally decreases, $-m'(q^\star) < 0$.*

MSLD says that consumers become more sensitive to percentage price changes as the price rises along a demand curve. As a firm expands output and lowers its price, demand therefore becomes less elastic, allowing a higher markup. The firm's first-order condition gives the markup $\mu(q) = [1 + \varepsilon(q)]^{-1}$, which increases with quantity under MSLD. Lower-cost firms operating further down a common demand curve thus charge higher markups.[16] Cost reductions are partly absorbed in higher markups, while cost increases compress them, dampening pass-through into prices (Mrázová and Neary, 2017; Melitz, 2018).

MSLD* extends the formalization of this same economic idea to the elasticity of marginal revenue. The two restrictions are closely linked: if the absolute elasticity of marginal revenue increases globally, then MSLD follows as well (Mayer et al., 2021; Melitz, 2018). The anti-laws describe the economic opposite: consumers become less sensitive to percentage price changes at higher prices, with anti-MSLD* also reversing the marginal-revenue condition. The literature generally

[16] For the role of demand in selection and sorting among heterogeneous firms, see Matsuyama and Ushchev (2022).

treats MSLD and MSLD* as the empirically relevant cases, consistent with evidence on incomplete cost pass-through, higher markups among more productive firms, and sales reallocations toward better-performing products (Melitz, 2018; Mayer et al., 2021). We focus on these reinforcing patterns, setting aside locally conflicting signs that capture neither the law nor its economic opposite.

Policy curvature. A policy's slope measures responsiveness to a cost shock; its curvature relative to slope measures response asymmetry. For any twice differentiable log policy Y, define

$$\rho[Y](\epsilon) := \frac{Y''(\epsilon)}{|Y'(\epsilon)|}. \tag{10}$$

Negative values indicate concavity; more negative values indicate greater concavity relative to slope. At the no-shock benchmark $\epsilon^\star = 0$, write $\rho^\star_Y := \rho[Y](0)$, with $Y = \mathcal{Q}$ for output and $Y = \mathcal{R}$ for real sales. Since $\mathcal{Q} = H^{-1}$, Appendix Lemma 2 gives

$$\rho^\star_{\mathcal{Q}} = \frac{\mathcal{Q}''(0)}{|\mathcal{Q}'(0)|} = \frac{H''(\log q^\star)}{H'(\log q^\star)^2}.$$

We can therefore read policy curvature from the first-order condition. For the price-taking benchmark, H is affine and $\rho^\star_{\text{pt}} = 0$.

Output responses. The following proposition expresses the output policy's slope and normalized curvature in terms of the elasticity of marginal revenue.

Proposition 2 (Output policy on a given inverse-demand curve)**.** *Define the* marginal cost–revenue elasticity gap *as $D(q) := \eta - 1 - m(q)$, the difference between the elasticities of marginal cost and marginal revenue. Then $D(q^\star) > 0$ and*

$$\mathcal{Q}'(0) = -\frac{1}{D(q^\star)}, \qquad \rho^\star_{\mathcal{Q}} = \frac{q^\star m'(q^\star)}{D(q^\star)^2}. \tag{11}$$

Consequently, MSLD implies a locally concave output policy, $\rho^\star_{\mathcal{Q}} < 0$, whereas anti-MSLD* implies a locally convex output policy, $\rho^\star_{\mathcal{Q}} > 0$.*

The policy slope is negative because an increase in costs reduces optimal output. Its magnitude, $1/D(q^\star)$, measures the strength of this response: a larger marginal cost–revenue elasticity gap implies a flatter output policy. Policy curvature describes how the response changes with the shock. Under MSLD*, a favorable cost shock moves the firm to a higher-quantity region in which marginal revenue is more elastic. The firm then absorbs part of the cost reduction through a higher markup

rather than expanding output proportionally. An adverse cost shock moves the firm toward a lower-quantity region in which marginal revenue is less elastic; as a result, the output policy is steeper in the cost shock. The resulting output policy is locally concave.

We now compare a monopolistic ("large") firm with a price-taking ("small") firm within the same framework. This comparison provides an initial benchmark for relating firm size to policy shape. Corollary 1 shows that price setting flattens the policy and, under MSLD*, introduces negative curvature.

Corollary 1 (Price-taking benchmark)**.** *The price-taking output policy satisfies* $\mathcal{Q}'_{\text{pt}}(0) = -1/(\eta - 1)$ *and* $\rho^\star_{\text{pt},\mathcal{Q}} = 0$*. A price-setting firm satisfies* $|\mathcal{Q}'(0)| < |\mathcal{Q}'_{\text{pt}}(0)|$ *and, under MSLD***,* $\rho^\star_{\mathcal{Q}} < \rho^\star_{\text{pt},\mathcal{Q}} = 0$*.*

Revenue responses. We now extend the analysis to real revenue, whose growth is our empirical outcome. The output and conditional real-revenue policies are linked by $\mathcal{R}(\epsilon) = \mathcal{Q}(\epsilon) + \log p(e^{\mathcal{Q}(\epsilon)})$. This identity lets us obtain the revenue policy's slope and curvature directly from the output policy and inverse demand.

Proposition 3 (Real-revenue-policy slope and curvature)**.** *Suppose* $p \in C^2$*,* $p'(q) < 0$*,* $\varepsilon^\star \in (-1, 0)$*, and* $\mathcal{Q}'(0) \neq 0$*. Then the real-revenue policy has slope and curvature index*

$$\mathcal{R}'(0) = (1 + \varepsilon^\star)\mathcal{Q}'(0) = -\frac{1 + \varepsilon^\star}{D(q^\star)}, \tag{12}$$

$$\rho^\star_{\mathcal{R}} = \rho^\star_{\mathcal{Q}} + \frac{\varepsilon^\star \psi^\star}{1 + \varepsilon^\star} \left|\mathcal{Q}'(0)\right|. \tag{13}$$

Because $1+\varepsilon^\star \in (0, 1)$*, relative-price adjustment makes the real-revenue policy flatter than the output policy. Under MSLD,* $\psi^\star > 0$*, so the relative-price contribution to curvature is negative. Under MSLD***, both curvature contributions are negative, implying* $\rho^\star_{\mathcal{R}} < 0$*. Under anti-MSLD***, both contributions are positive, implying* $\rho^\star_{\mathcal{R}} > 0$*. The price taker satisfies* $\mathcal{R}'_{\text{pt}}(0) = \mathcal{Q}'_{\text{pt}}(0)$ *and* $\rho^\star_{\text{pt},\mathcal{R}} = 0$*.*

An output adjustment along a downward-sloping demand curve induces an offsetting price movement. This makes real revenue respond less strongly than physical output. Curvature contains two components: curvature of the output policy and curvature introduced directly by relative-price adjustment. Under MSLD, demand becomes less elastic as the firm expands, so price declines offset an increasing share of quantity gains in revenue terms. This adds negative curvature to the real-revenue policy.

3.3.3 Firm Size and Policy Shape

The preceding analysis takes a firm's baseline operating point $q^\star = \exp(\mathcal{Q}(0))$ as given. We now compare firms that face the same inverse-demand function but operate at different baseline quantities because their baseline productivity differs.

Index these baseline operating points by q. The associated cost shifter is $\phi_q := R'(q)/(\eta q^{\eta-1})$, or, equivalently, productivity is $1/\phi_q$. By construction, q solves this firm's problem at $\epsilon = 0$. Let $\mathcal{Q}_q$ and $\mathcal{R}_q$ denote its log-output and log-real-revenue policies. For $Y \in \{\mathcal{Q}, \mathcal{R}\}$, define the policy slope and normalized curvature at baseline size q by

$$\beta_Y(q) := Y_q'(0), \qquad \rho_Y(q) := \frac{Y_q''(0)}{|Y_q'(0)|}.$$

We trace how these coefficients vary with firm size, starting with firms that produce small quantities. Suppose inverse demand has a finite positive choke price, $p(0) = \bar{p} \in (0, \infty)$. These firms operate near that price. Proposition 4 shows that, if inverse demand leaves the choke price with a finite negative slope, absolute policy slopes decline with operating quantity while normalized curvature becomes more negative.

Proposition 4 (Output policy shape near a finite choke price)**.** *Suppose $p \in C^4([0, \bar{q}))$ for some $\bar{q} > 0$, $p(0) = \bar{p} \in (0, \infty)$, and $p'(0) \in (-\infty, 0)$. Then, as $q \downarrow 0$,*

$$\beta_{\mathcal{Q}}(q) = -\frac{1}{\eta - 1} + C_{\mathcal{Q}} q + O(q^2), \quad \rho_{\mathcal{Q}}(q) = -C_{\mathcal{Q}} q + O(q^2), \quad C_{\mathcal{Q}} := \frac{-2p'(0)}{\bar{p}(\eta-1)^2} > 0. \tag{14}$$

Consequently, for all sufficiently small $q > 0$,

$$\beta_{\mathcal{Q}}'(q) > 0, \qquad \rho_{\mathcal{Q}}'(q) < 0, \qquad \frac{d|\beta_{\mathcal{Q}}(q)|}{dq} < 0, \qquad \frac{d|\rho_{\mathcal{Q}}(q)|}{dq} > 0.$$

Under the boundary condition $p'_+(0) \in (-\infty, 0)$, firms producing near the choke price can make accurate demand and revenue calculations using the local slope of inverse demand. Let $\delta > 0$ denote a small price concession below $\bar{p}$. Locally, $\bar{p} - p(q) \approx -p'_+(0)q$. Setting $p = \bar{p} - \delta$ therefore implies $q \approx \delta/[-p'_+(0)]$ and $pq \approx \bar{p}\delta/[-p'_+(0)]$. Thus, output and revenue initially increase proportionally as the price falls below the choke price. In contrast, if $p'_+(0) \in \{0, -\infty\}$, these linear extrapolations are not informative.

The boundary calculation in the proof of Proposition 4 (see Appendix, Equation 28) shows that $p'_+(0) < 0$ is equivalent to the two inequalities defining MSLD*.

By continuity, MSLD* holds for all sufficiently small operating quantities. The boundary expansions therefore establish a comparison between firms facing the same inverse-demand curve but differing in baseline costs: near the choke price, lower-cost firms operate at larger quantities and have flatter, more concave output policies.

Proposition 4 concerns physical output. We next establish that real-revenue policies exhibit the same local decline in absolute slopes and increase in normalized curvature magnitude with firm size. Because marginal revenue is positive near the choke price, total revenue is locally increasing in output. Firms can therefore be ranked by real revenue as well as by physical quantity.

Proposition 5 (Real-revenue policy shape near a finite choke price)**.** *Under the hypotheses of Proposition 4, as* $q \downarrow 0$*,*

$$\beta_{\mathcal{R}}(q) = \frac{-1}{\eta - 1} + C_R q + O(q^2),\ \rho_{\mathcal{R}}(q) = -C_R q + O(q^2),\ C_R := \frac{-p'(0)}{\bar{p}} \frac{\eta + 1}{(\eta - 1)^2} > 0. \quad (15)$$

Consequently, for all sufficiently small $q > 0$*,*

$$\beta'_{\mathcal{R}}(q) > 0, \qquad \rho'_{\mathcal{R}}(q) < 0, \qquad \frac{d|\beta_{\mathcal{R}}(q)|}{dq} < 0, \qquad \frac{d|\rho_{\mathcal{R}}(q)|}{dq} > 0.$$

Because $d[p(q)q]/dq = R'(q) > 0$*, these are gradients in revenue size and in physical quantity, too.*

As baseline output approaches zero, both log policies approach those of the price taker locally: their slopes converge to $-1/(\eta - 1)$ and their normalized curvatures to zero.

Propositions 4 and 5 use a finite, nonzero slope at the choke price as a transparent sufficient condition. Appendix Remark 2 extends these relationships between firm size, policy slope, and curvature to $p(q) = \bar{p} - cq^{\gamma} + o(q^{\gamma})$, with $c, \gamma > 0$, allowing inverse demand to be flat or infinitely steep at the boundary. This is the local form of the generalized-linear inverse-demand family introduced by Bulow and Pfleiderer (1983). For example, the Klenow–Willis form of Kimball demand need not have a finite, nonzero slope at the choke price, yet satisfies this more general condition for every positive Kimball curvature parameter (Klenow and Willis, 2016).

Policy Differences and Markups. We have shown in two ways how market power makes larger firms less responsive but more asymmetric. First, comparing a price-taking firm with a price-setting firm provides a benchmark: under MSLD*, price setting dampens responses to cost shocks but makes them more asymmetric.

Second, we establish how policy slopes and curvature vary continuously with firm size among price-setting firms facing a common inverse-demand curve. Near the choke price, more productive firms operate at larger quantities and have flatter policies with greater curvature magnitude.

Within this model, the demand properties that shape transmission also govern markups. Under MSLD, lower-cost firms operating at larger quantities face less elastic demand and charge higher markups. Such markup differences can distort allocation (Dhingra and Morrow, 2019; Edmond et al., 2023). CES, the efficient benchmark in Dhingra and Morrow (2019), provides the contrast: log policies are linear and markups constant, eliminating this demand-induced asymmetry and markup dispersion.

The next section derives the growth moments implied by these policies and assesses alternative explanations of the stylized facts.

4 Growth Moments and Their Interpretation

We now link the model to the empirical patterns documented in Section 2. We first derive firms' growth moments, then show how the policy shapes established in Section 3 account for their cyclical movements and opposing size gradients. Finally, we examine alternative explanations.

4.1 Growth Moments

We now introduce a date index t for the equilibria above. Firms have no capital or other intertemporal choices, so their optimization problems are separable across dates: the sequence of static equilibria is also the dynamic equilibrium of the model. Exposures remain fixed, while aggregate and idiosyncratic shocks vary across dates. The idiosyncratic shocks κ_{it} are iid across firms and time. We condition on u_t without specifying its distribution.[17]

Policy Representation. Consider firms within a group sharing the same real-revenue policy, and suppress the group index. Such groups can differ in baseline productivity and hence operating size (see Remark A.1 in Online Appendix A).

[17]For cross-sectional growth moments at a given date, we allow any nonnegative adapted process for u_t. To study how these moments vary over time, we additionally require stationarity, ergodicity, and the relevant finite moments. One example is the stationary shot-noise process $u_t = \beta u_{t-1} + N_t$, where $N_t \overset{\text{iid}}{\sim} \text{Poisson}(\omega)$ and $\beta \in (0, 1)$: innovations raise costs, and their effects decay geometrically.

Writing $b_R = \bar{\lambda}/\eta$, first-differencing the equilibrium real-revenue policy in Equation (5) gives growth:[18]

$$g_{it} = \mathcal{R}(\kappa_{it} + \lambda_i u_t) - \mathcal{R}(\kappa_{i,t-1} + \lambda_i u_{t-1}) - b_R \Delta u_t + \Delta\zeta_{it}. \tag{16}$$

Without heterogeneous exposure or idiosyncratic shocks, all firms therefore have the same real-sales growth, regardless of baseline productivity. With heterogeneous shocks, differences in firms' effective shocks, transmitted through $\mathcal{R}$, determine the leading variance and skewness.

Under the symmetry conditions described in Section 3.2, the model implies $\zeta_{it} = Z_t + \nu_{it}$, where Z_t is common across firms and ν_{it} is third order in the shock scale. Thus $-b_R\Delta u_t$ and ΔZ_t shift mean growth, while heterogeneity in the price residual is negligible at the orders considered below.

The remaining heterogeneity enters through the centered shocks ϵ_{it}, which firms transmit through $\mathcal{R}$. For small shocks, we can therefore characterize the growth distribution using the local slope $\mathcal{R}'(0)$ and normalized curvature $\rho^\star_{\mathcal{R}}$. Proposition 6 expresses the mean, variance, and skewness of cross-sectional growth, and their variation over time, in terms of these coefficients.

Proposition 6 (Approximate Conditional and Time-Series Growth-Rate Moments)**.** *Let $\mathcal{F}_t$ denote the aggregate information available at time t. Let all required moments be finite, let λ be symmetrically distributed, and let the idiosyncratic shocks κ_{it} follow the time-invariant symmetric distribution specified above. If real-sales growth is given by Equation (16) and the price residual is common across firms up to a third-order firm-specific remainder, the following statements hold up to the stated approximation orders in a common local scale s_t of the centered shocks $\epsilon_{it} = \kappa_{it} + \lambda_i u_t$ at dates t and $t-1$.*[19]

i) The conditional cross-sectional moments are

$$\mu_t := \mathbb{E}[g_{it} \mid \mathcal{F}_t] = -b_R\Delta u_t + \frac{\mathcal{R}''(0)}{2}\Delta(u_t^2)\sigma_\lambda^2 + \Delta Z_t + O(s_t^3), \tag{17}$$

$$v_t := \mathbb{V}(g_{it} \mid \mathcal{F}_t) = \mathcal{R}'(0)^2\left((\Delta u_t)^2\sigma_\lambda^2 + 2\sigma_\kappa^2\right) + O(s_t^3), \tag{18}$$

$$\Gamma_t := \gamma_1(g_{it} \mid \mathcal{F}_t) = \rho^\star_{\mathcal{R}}\frac{3}{2}\frac{4(\Delta u_t)(u_{t-1}+u_t)\sigma_\kappa^2\sigma_\lambda^2 + \Delta(u_t^2)(\Delta u_t)^2\mathbb{V}(\lambda^2)}{\left(2\sigma_\kappa^2 + (\Delta u_t)^2\sigma_\lambda^2\right)^{3/2}} + O(s_t^2). \tag{19}$$

[18] For clarity, we omit deterministic trends. Adding at to log real sales adds a to growth. This common shift affects mean growth but leaves cross-sectional variance and skewness unchanged.

[19] Here, $s_t \downarrow 0$ denotes a common local scale such that both ϵ_{it} and $\epsilon_{i,t-1}$ are $O(s_t)$, uniformly across firms. We write $X_t = O(s_t^k)$ when X_t/s_t^k remains bounded as $s_t \downarrow 0$, and $X_t = \Theta(s_t^k)$ when $|X_t|/s_t^k$ is bounded above and away from zero. For random quantities, these bounds apply in the norm relevant to the stated moment.

ii) Suppose additionally that the aggregate process $\{u_t\}_t$ is stationary and ergodic. Holding the shock and exposure processes fixed,

$$\begin{aligned} \mathbb{E}_{\text{time}}[v_t] &\propto \mathcal{R}'(0)^2, & \text{SD}_{\text{time}}(v_t) &\propto \mathcal{R}'(0)^2, \\ \mathbb{E}_{\text{time}}[\Gamma_t] &\propto \rho^\star_{\mathcal{R}}, & \text{SD}_{\text{time}}(\Gamma_t) &\propto |\rho^\star_{\mathcal{R}}|. \end{aligned} \tag{20}$$

The proposition also holds for output growth, replacing $\mathcal{R}$ and its coefficients with their $\mathcal{Q}$-counterparts and setting $\zeta_{it} = 0$.

Proof. See Online Appendix B.1.

4.2 Implications for the Stylized Facts

Assume that the results of Section 3.3 apply. In particular, Marshall's Strong Second Law of Demand holds, so real-revenue policies are concave, and the conditions for the local size comparison are satisfied. As baseline firm size increases, policies therefore become flatter and normalized curvature becomes more negative. With these properties in place, we use Proposition 6 to interpret the stylized facts through the model.

Mean and Stylized Fact 1. Equation (17) separates average real-sales growth into three effects. First, when u_t rises, the common cost component increases and $-b_R \Delta u_t$ lowers average growth. Second, heterogeneous exposure spreads firms out along the concave policy $\mathcal{R}$: firms receiving favorable shocks expand less than firms receiving equally large adverse shocks contract. This Jensen effect further lowers average growth and is captured by the term involving $\mathcal{R}''(0)$. Both effects reverse as the aggregate cost factor recedes.

Finally, ΔZ_t captures the change in the common proportional price forecast error. It raises measured real-sales growth when realized prices exceed conjectured prices by a larger proportion than in the previous period, and lowers growth when that proportion falls. Because this adjustment is common across firms up to second order, it affects mean growth but leaves the leading variance and skewness unchanged.[20]

Dispersion and Stylized Fact 2. With fixed exposures λ_i, the aggregate contribution to growth variance in Equation (18) depends on $(\Delta u_t)^2$: firms contract

[20]We expect ΔZ_t to be small. The finding that oblivious equilibria can closely approximate full-information Markov-perfect equilibria motivates this expectation (Weintraub et al., 2008). If the remaining price discrepancy is persistent, taking differences may reduce its contribution further.

by different amounts when u_t rises and recover by different amounts when it falls. Larger movements in either direction therefore increase dispersion, consistent with the V-shaped pattern around average growth in Figure 2. Idiosyncratic shocks contribute a constant component to growth variance.

Across size groups, both average growth variance and its time-series standard deviation scale with $\mathcal{R}'(0)^2$. The flatter policies of larger firms therefore generate lower and less variable growth dispersion, as in Stylized Fact 2.

Skewness and Stylized Fact 3. Equation (19) gives the corresponding skewness pattern. When an adverse aggregate shock arrives, the dispersion of $\lambda_i u_t$ increases. The concave policy maps this wider shock incidence into disproportionately large contractions, so growth skewness turns negative as mean growth falls. During recovery, the firms that contracted most rebound most strongly: mean growth recovers and skewness turns positive. Changes in the aggregate shock are essential. If $u_t = u_{t-1}$, the leading skewness term vanishes, even if the cost factor remains elevated.

The more negative normalized curvature of larger firms amplifies both sides of the cycle: their growth skewness becomes more negative during contractions and more positive during recoveries. Its time-series standard deviation scales with $|\rho^\star_{\mathcal{R}}|$, generating the larger skewness fluctuations in Stylized Fact 3. Together with the slope gradient, this explains why larger firms have less variable growth dispersion but more variable skewness.

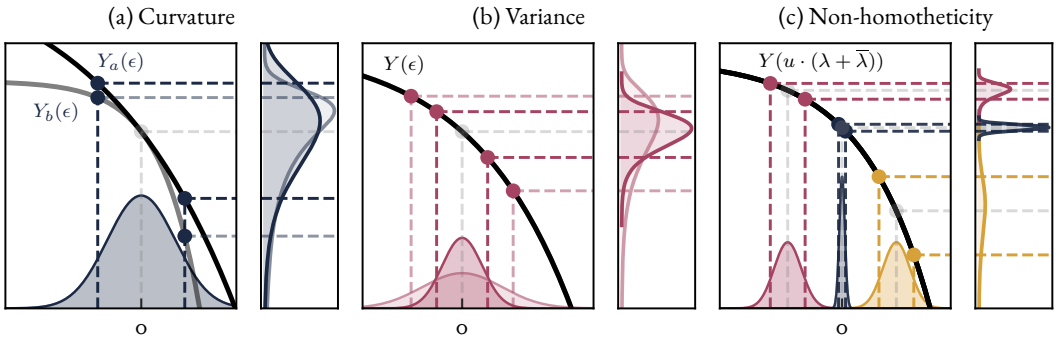
An aggregate shock that decays geometrically gives a simple illustration. Suppose the assumptions of Proposition 6 hold, there are no idiosyncratic shocks, and the aggregate factor follows the path $(u_0, u_1, u_2, \ldots) = x \cdot (0, \beta^0, \beta^1, \beta^2, \ldots)$, with $x > 0$ and $\beta \in (0, 1)$. The moment approximation yields the following result.

Corollary 2 (Moment impact and rebound)**.** *Consider the preceding shock path with $\sigma^2_\kappa = 0$ and $\rho^\star_{\mathcal{R}} < 0$. To leading order, mean growth and skewness are negative on impact, while growth dispersion rises. During recovery, mean growth μ_t, growth dispersion $\sqrt{v_t}$, and skewness Γ_t are all positive and decay by a factor of β each period.*

Under the restrictions on policy slopes and curvature established in Section 3.3.3, and with common shock and exposure processes, larger firms have lower growth variance but greater absolute skewness on impact and throughout recovery.

Proof. See Online Appendix B.1.

Figure 4: Skewness in Firm Growth: Concavity, Shock Variance, and Non-Homotheticity



Note: Panel (a) maps symmetric shock incidence into skewed outcomes through a concave policy. Greater local concavity under policy b produces more left skewness. Panel (b) holds the policy fixed and increases shock variance, amplifying outcome skewness. Panel (c) illustrates non-homotheticity: aggregate states shift the shock distribution, changing the slopes and curvatures at which firms operate. Colors distinguish states; dashed lines mark the 10th, 50th, and 90th percentiles.

Figure 4(a–b) illustrates the two ingredients behind this skewness response. Panel (a) holds the shock distribution fixed: greater policy concavity produces more left-skewed levels of the outcome variable, whether log revenue or log output. Panel (b) holds the policy fixed: increasing shock dispersion without changing its mean also amplifies this asymmetry. The panels illustrate mappings into levels; their implications for growth follow from comparing each firm's policy outcomes across dates, as in Equation (16).

Discussion. These joint predictions depend on demand shape. Log-linear policies generate no skewness through this transmission channel. Anti-MSLD* implies convex policies and countercyclical skewness; along a common inverse-demand curve, it also makes policies steeper at larger operating quantities. Both predictions run counter to the evidence. The joint patterns therefore support MSLD* within our model and provide independent aggregate evidence for Marshall's Second Law, alongside evidence from sales reallocation (Mayer et al., 2021) and price and markup adjustment (Berman et al., 2012; Amiti et al., 2019). In our model, markup adjustments generate the asymmetry and opposing size gradients. These patterns therefore reveal variable market power across firms, a source of misallocation (Dhingra and Morrow, 2019).

Our results rely on separating the mean effect of the common cost change, $-b_R \Delta u_t$, from the nonlinear term $\Delta \mathcal{R}(\epsilon_{it})$ in Equation (16). Homothetic demand and the equilibrium conjectures permit this separation around a fixed no-shock

benchmark. Under non-homothetic demand, the common aggregate effect can instead remain inside the policy, shifting the location at which firms evaluate it, as Figure 4(c) illustrates. This location channel also appears in the growth-response maps $g_{it} = f(s_{it})$ of Ilut et al. (2018) and Straub and Ulbricht (2019). The separation in our model makes Proposition 6 useful for empirical work: policy differences can be characterized by local slopes and curvatures at a fixed benchmark. Because the moment responses arise through transmission rather than primitive higher-order shocks, the growth distribution reveals features of firms' policies and, through the model, demand curvature.

Remark 1 (Revenue productivity)**.** *The distinction between shocks and transmission also applies to revenue-based productivity measures (Salgado et al., 2026; Ilut et al., 2018). With production $q_i = A_i X_i^{1/\eta}$, revenue productivity satisfies*

$$\mathrm{TFPR}_i = \frac{r_i}{X_i^{1/\eta}} = \frac{p_i}{P} A_i,$$

and therefore includes an endogenous relative price. Under the regular finite-choke-price conditions, its conditional log policy is locally concave.[21] *The preceding growth-moment argument then implies procyclical skewness in TFPR growth despite symmetric centered shocks to log physical productivity. Revenue-productivity skewness therefore need not originate in technology.*

4.3 Alternative Mechanisms

Could differences in the shocks firms receive explain the same facts without differences in transmission?[22] Could technology generate the required policy differences? The joint behavior of variance and skewness restricts both alternatives. For empirical size groups, we maintain the price-residual bound within each group, without treating a size decile as a single structural type.

There is a trade-off between the structure of transmission and that of shocks.

[21] Let X_i be a composite input with fixed common unit cost normalized to one. Setting $A_i^{-\eta} = \phi_i e^{\lambda u + \epsilon_i}$ reproduces the model's cost function. After separating common terms and the price residual, the conditional log policy is, up to a constant, $\mathcal{T}(\epsilon) = \mathcal{R}(\epsilon) - \mathcal{Q}(\epsilon) - \epsilon/\eta$. Writing $p(q) = \bar{p}[1 - cq + O(q^2)]$, with $c > 0$, the boundary expansions give

$$\mathcal{T}''(0) = \mathcal{R}''(0) - \mathcal{Q}''(0) = -\frac{cq^\star}{(\eta-1)^2} + O\big((q^\star)^2\big) < 0$$

for sufficiently small baseline quantity $q^\star > 0$.

[22] Related work studies cyclical productivity dispersion (Kehrig, 2015), endogenous firm volatility through market exposure (Decker et al., 2016), and customer diversification (Herskovic et al., 2020).

With linear transmission, matching the stylized facts requires effective shock distributions whose dispersion and skewness vary both over time and across firm sizes. Simpler firm policies therefore require richer assumptions about the shocks firms receive. Allowing these distributions to vary freely to match each group's growth moments effectively assumes the patterns we seek to explain. Purely linear transmission also runs against the evidence of Ilut et al. (2018), who estimate concave employment-growth responses to productivity shocks. We maintain a common, symmetric exposure distribution and recover differences in policy shape, while allowing idiosyncratic risk to vary with size.

Exposure Distribution. Could greater exposure dispersion among smaller firms explain the opposing size gradients? Hold policies and the aggregate shock process fixed across groups, and suppose idiosyncratic risk is negligible. If large firms' centered exposures are obtained by scaling small firms' exposures by $d \in (0, 1)$, Equations (18) and (19) imply

$$v_{L,t} \simeq d^2 v_{S,t}, \qquad \Gamma_{L,t} \simeq d\Gamma_{S,t}.$$

Less dispersed exposure therefore lowers both growth dispersion and the amplitude of skewness fluctuations. The first prediction matches the data; the second goes in the opposite direction.

Therefore, an exposure-based explanation must involve differences in distributional shape. Larger firms would need lower exposure variance but a greater skewness-amplification factor $C_\lambda := (3/2)\,\mathrm{Var}(\lambda^2)/\sigma_\lambda^3$ (setting $\sigma_\kappa = 0$ in both groups, this factor enters Equation (19) multiplicatively alongside $\rho^\star_{\mathcal{R}}$). The COVID and inflation exposure proxies of Davis et al. (2025) provide a direct check on these required shape differences.[23] The exposure moments are non-monotonic across size deciles, and their confidence intervals are consistent with equal values across deciles. Appendix D.5 reports the construction and results in Figure D.6.

Idiosyncratic Shocks. Equation (18) shows that smaller firms can have more dispersed growth because their policies are steeper, their aggregate-shock exposures are more dispersed, or their idiosyncratic shocks have larger variance. Greater idiosyncratic risk also attenuates standardized skewness, so it can contribute to

[23]This evidence is limited to two shock types during 2020–2022. Moreover, COVID was an unusually large disturbance, so the local moment approximations in Proposition 6 may be less accurate for that episode.

smaller firms' less variable skewness even under identical policy curvature.[24]

We address this possibility in two ways. First, time-invariant idiosyncratic risk adds a constant to growth variance and therefore does not change its time-series standard deviation at the retained approximation order. It cannot by itself explain the declining variability of dispersion with firm size in Figure 3. Second, Section 5.2 jointly identifies relative policy slopes and curvature magnitudes while explicitly allowing σ_κ to vary across size deciles. After accounting for these differences, we still find flatter policies and greater curvature magnitudes among larger firms.

Alternative Sources of Policy Curvature. Recovering policy slopes and curvature does not require a demand-side explanation of these policies. Technological heterogeneity provides another possible source, but it must meet three requirements. First, it must reproduce declining absolute policy slopes and increasing curvature magnitudes with firm size. Second, these gradients must persist within industries. The moment gradients are similar under within-industry and economy-wide size rankings (Appendix Figure D.2); Section 5.2 shows the same robustness for the recovered policy slopes and curvature coefficients. Third, it must go beyond differences in isoelastic cost exponents, which leave marginal costs log-linear in output and cannot supply the required curvature. A technological explanation must therefore account for how cost curvature changes with firm size. More complex production-side mechanisms can generate asymmetric responses (Miranda-Pinto et al., 2023; Dew-Becker and Vedolin, 2025; Kamepalli et al., 2025). These papers do not, however, establish the size gradients documented here: lower and less variable growth dispersion alongside greater fluctuations in skewness among larger firms.

Our demand framework delivers these joint policy restrictions from preferences, heterogeneous incidence, and profit maximization, without additional adjustment or financing frictions. The next section recovers the policy differences, tests the size gradients using identified shocks, and examines whether greater measured exposure dispersion predicts stronger distributional responses around COVID.

5 Empirical Evidence

We assess the theory in three complementary exercises. First, we recover relative policy responsiveness, normalized curvature magnitude, and idiosyncratic volatility from unconditional growth moments, without imposing the demand-shape results.

[24] For a time-varying uncertainty mechanism based on news about future volatility, see Berger et al. (2020).

Second, we ask whether responses to identified aggregate shocks reproduce the opposing size gradients. Third, we test whether more dispersed measured exposure predicts stronger distributional changes around COVID.

5.1 Data and Sample

We use the Compustat sample introduced in Section 2, covering U.S. public firms over 1984Q1–2024Q4. As stated before, real sales are nominal sales deflated by the GDP deflator, and growth is the year-on-year change in log real sales. Firm size is average real sales over the preceding four quarters, and size comparisons are made within two-digit NAICS industries. We require positive reported sales in the quarters needed to construct growth and size, and trim the top and bottom 1 percent of growth observations each quarter. Appendix C details sample construction.

Compustat firms are larger than the representative manufacturing firms in the QFR comparison sample (Appendix Table C.1). Our size comparisons therefore concern heterogeneity within the upper part of the U.S. firm-size distribution: even firms classified as small in our sample need not be small in the broader economy. The following subsections describe additional sample requirements and size classifications used in each exercise.

5.2 Recovering Policy Differences

Larger firms have lower growth dispersion but greater skewness fluctuations. Lower idiosyncratic risk can amplify standardized skewness fluctuations even when policy curvature is unchanged. We use growth variance and skewness jointly to distinguish differences in firms' policies from differences in their idiosyncratic risk. Throughout, we maintain a common symmetric exposure distribution across size groups and allow idiosyncratic risk to differ across groups but remain constant over time, as discussed in Section 4.3. The curvature index measures the second derivative of the revenue policy relative to its absolute slope. Because our exercise compares the amplitude of adjusted skewness fluctuations, it identifies relative curvature magnitudes, but not their signs.

Identification. We use the baseline within-industry size deciles, pooling firms with the same rank across industries. Let v_{jt} and Γ_{jt} denote growth variance and cumulant skewness in decile j, and let $m_t := \mu_t - \mathbb{E}_{\text{time}}[\mu_t]$ denote mean growth across all firms, demeaned over time. Policy slopes, idiosyncratic volatility, and curvature magnitudes are reported relative to Decile 5.

Policy slopes. We first use fluctuations in growth variance to identify differences in policy responsiveness. At the retained approximation order, time-invariant idiosyncratic risk adds the same component to variance in every period, so it does not affect these fluctuations. With a common exposure distribution and aggregate shock process, their amplitude scales with the squared policy slope. Taking the square root of the ratio of variance fluctuations across groups therefore identifies relative absolute policy slopes, even when idiosyncratic risk differs.

Idiosyncratic risk. We next use variance levels to distinguish greater idiosyncratic risk from steeper policies. Equation (17) gives $\Delta u_t = -m_t/b_m + O(s^2)$, where $b_m > 0$ is the average coefficient on $-\Delta u_t$ in the group mean equations. Substituting this into the variance equation yields $v_{jt} \simeq \alpha_j + \beta_j m_t^2$. The intercept α_j captures the idiosyncratic component of growth variance, while $\beta_j m_t^2$ captures the aggregate component. Both scale with the squared policy slope. Dividing α_j by β_j therefore removes policy responsiveness and gives idiosyncratic variance up to a common scale. Comparing these ratios across groups and taking the square root then identifies relative idiosyncratic volatility. To implement this comparison, we construct α_j and β_j from the time-series moments defined below.

Policy curvature. We then adjust skewness for idiosyncratic risk to identify relative curvature magnitude. To leading order, Equation (19) can be written as

$$\Gamma_{jt} \simeq \rho^{\star}_{\mathcal{R},j} C_\lambda (u_t + u_{t-1}) \operatorname{sgn}(\Delta u_t) A_{jt},$$

where C_λ summarizes how the common exposure distribution amplifies skewness, and A_{jt} is an adjustment factor for idiosyncratic risk, defined in the proposition below. Dividing skewness by A_{jt} leaves a common aggregate factor multiplying each group's curvature. Comparing the root mean squares of these adjusted series on common dates cancels that factor and identifies relative curvature magnitude. Root mean squares measure the size of skewness movements without letting positive and negative values cancel.

Proposition 7 summarizes these relationships. Approximation errors use the local shock scale s.[25]

Proposition 7 (Relative identification from observable growth moments)**.** *Suppose the conditions of Proposition 6 hold within each group, together with the conditions of either Proposition A.1 or Remark A.1 in Online Appendix A. Assume a common*

[25] Define $\sigma_{\epsilon,j\tau} := (\sigma^2_{\kappa,j} + u^2_\tau \sigma^2_\lambda)^{1/2}$, $s_t := \max_{j,\,\tau\in\{t-1,t\}} \sigma_{\epsilon,j\tau}$, and $s := \sup_{t\in\mathcal{T}} s_t$ on the common support $\mathcal{T}$.

Λ *and time-invariant* $\sigma^2_{\kappa,j}$*. Define*

$$\beta_j := \frac{\mathrm{SD}_{\mathrm{time}}(v_{jt})}{\mathrm{SD}_{\mathrm{time}}(m_t^2)}, \qquad \alpha_j := \mathbb{E}_{\mathrm{time}}[v_{jt}] - \beta_j \mathbb{E}_{\mathrm{time}}[m_t^2].$$

Then the following quantities are identified relative to Decile 5:

$$q_j := \frac{|\mathcal{R}'_j(0)|}{|\mathcal{R}'_5(0)|} = \left[\frac{\mathrm{SD}_{\mathrm{time}}(v_{jt})}{\mathrm{SD}_{\mathrm{time}}(v_{5t})}\right]^{1/2} + O(s), \tag{21}$$

$$r_j^{\kappa} := \frac{\sigma_{\kappa,j}}{\sigma_{\kappa,5}} = \left(\frac{\alpha_j \beta_5}{\beta_j \alpha_5}\right)^{1/2} + O(s), \tag{22}$$

$$r_j^{\rho} := \frac{|\rho^{\star}_{\mathcal{R},j}|}{|\rho^{\star}_{\mathcal{R},5}|} = \frac{\mathrm{RMS}_{\mathrm{time}}(\Gamma_{jt}/A_{jt})}{\mathrm{RMS}_{\mathrm{time}}(\Gamma_{5t}/A_{5t})} + O(s). \tag{23}$$

Here $\mathrm{RMS}_{\mathrm{time}}(x_t) := \{\mathbb{E}_{\mathrm{time}}[x_t^2]\}^{1/2}$ *is the root mean square on the common support, and*

$$A_{jt} := \frac{|m_t|\,(2\alpha_j/\beta_j + \mathcal{K}_\lambda m_t^2)}{\mathcal{K}_\lambda\,(\alpha_j/\beta_j + m_t^2)^{3/2}}, \qquad \mathcal{K}_\lambda := \frac{\mathrm{Var}(\lambda^2)}{\sigma_\lambda^4}.$$

Proof. See Online Appendix B.2.

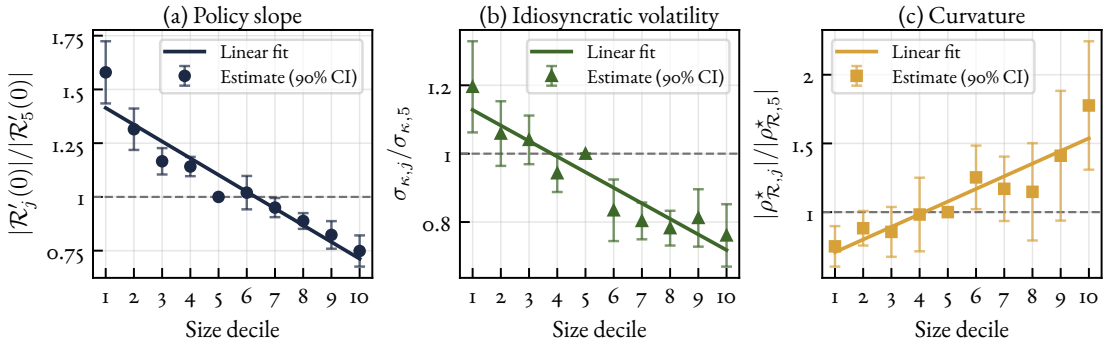
Although the common multiplier C_λ cancels from relative comparisons, identifying relative curvature requires specifying the exposure-shape coefficient $\mathcal{K}_\lambda$, which variance moments do not identify. Our baseline sets $\mathcal{K}_\lambda = 2$, the value implied by $\Lambda = \mathcal{N}(0, \sigma_\lambda^2)$. We assess sensitivity to this choice by varying $\mathcal{K}_\lambda$ over $\{1, 2, 4\}$, which has virtually no effect on our results (see also Appendix F). At $\mathcal{K}_\lambda = 2$, A_{jt}^2 has a transparent interpretation: it is the share of cross-sectional growth variance within size group j attributable to the aggregate shock.[26]

Estimation. We calculate quarterly moments in each size decile and retain quarters observed for all ten deciles. The baseline uses the 90 percent of these quarters with the smallest m_t^2, limiting the influence of extreme aggregate-growth observations on the local moment estimates. We estimate the quantities in Proposition 7 using their sample counterparts.

For the curvature adjustment only, we smooth the estimated ratios α_j/β_j across deciles using a positive fitted curve; the reported idiosyncratic-volatility estimates use the raw ratios. We also exclude dates on which the adjustment factor is close to

[26] Substituting $\mathcal{K}_\lambda = 2$ into the adjustment factor gives $A_{jt}^2 = m_t^2/(\alpha_j/\beta_j + m_t^2) = \beta_j m_t^2/(\alpha_j + \beta_j m_t^2)$. At the retained approximation order, α_j is the idiosyncratic contribution to growth variance and $\beta_j m_t^2$ is the aggregate contribution.

Figure 5: Relative Policy Slopes, Idiosyncratic Volatility, and Curvature Magnitude



Note: Within-industry size deciles; estimates normalized at Decile 5. Bars show 90% delta-method intervals using Newey–West covariance estimates with four calendar-quarter lags; lines are descriptive OLS fits.

zero, since division would magnify small errors in measured skewness.[27] Confidence intervals use the delta method with Newey–West covariance estimates, conditional on the selected dates and full-sample centering of mean growth. Appendix F details estimation, inference, and sensitivity to the exposure-shape and adjustment-factor choices.

Results. Figure 5 reports estimated policy slopes, idiosyncratic volatility, and curvature magnitudes across size deciles, each relative to Decile 5. The relative absolute policy slope falls from 1.58 in the smallest decile to 0.75 in the largest, while relative idiosyncratic volatility declines from 1.20 to 0.76. After accounting for this difference in idiosyncratic risk, relative curvature magnitude rises from 0.75 to 1.78. Larger firms therefore have substantially flatter policies but greater curvature.[28]

All three estimated patterns are also robust across different specifications. They persist when we retain 100, 90, 75, or 50 percent of balanced quarters ranked by m_t^2, and when we vary the exposure-shape coefficient and adjustment-factor cutoff (Appendix F). These patterns also persist with economy-wide size rankings and averages of estimates obtained separately by industry (Appendix F).

These estimates support the predicted decline in responsiveness and rise in normalized curvature magnitude with firm size. Their identification does not

[27] We fit $\widehat{\alpha}_j/(2\widehat{\beta}_j)$ by nonlinear least squares using $\exp(a+bj+cj^2+dj^3)$. Comparisons require $A_{jt} \geq 0.05$ in both deciles and at least eight qualifying quarters. Median adjustment factors are well above this cutoff. Profile correlations with the baseline are 0.877, 0.832, and 0.979 at cutoffs of 0.01, 0.025, and 0.075, respectively.

[28] Spearman's correlations and 90% parametric intervals are -0.988 $[-1.000, -0.952]$ for policy slopes, 0.939 $[0.709, 0.976]$ for curvature magnitude, and -0.952 $[-0.976, -0.758]$ for idiosyncratic volatility. Table F.1 reports the full rank-inference results.

require a demand microfoundation, so they also discipline models satisfying the moment framework's assumptions: responsiveness must decline and curvature magnitude must rise with firm size.

5.3 Responses to Identified Aggregate Shocks

Do the opposing size gradients also appear in firms' responses to identified aggregate shocks? We estimate firm-level responses to six innovations and compare their dispersion and skewness across size groups within industries.[29] At each shock's horizon of largest average contraction, the model predicts more dispersed responses among small firms and more negatively skewed responses among large firms.

Aggregate innovations and sample. We study innovations to the main business cycle (Angeletos et al., 2020), credit spreads (Gilchrist and Zakrajšek, 2012), oil supply news (Känzig, 2021), monetary policy (Jarociński and Karadi, 2020), investment news (Ben Zeev and Khan, 2015), and financial uncertainty (Ludvigson et al., 2021). Appendix E.1 documents their sources and controls. Each innovation is scaled by its fixed sample standard deviation and oriented so that positive values are growth-contracting. Samples start in 1984Q1, except monetary policy in 1994Q1, and end in 2019Q4, except the main business-cycle series in 2017Q4. Both shock origins and outcomes are restricted to these windows to limit the influence of COVID-19. We retain uninterrupted observation periods containing at least 40 quarters of usable growth for a given firm (a *spell*), allowing several such spells per firm. To classify the spell as a large-firm (small-firm) spell, the firm must remain in the top (bottom) half of its within-industry size distribution throughout the spell. Firm size is measured by average real sales over the preceding four quarters.

Firm-level responses. We estimate firm-level responses using an approach similar to Krusell et al. (2026). We allow responses to adverse and favorable innovations to differ, accommodating the asymmetric adjustment predicted by the mechanism. For firm-spell j, shock s, and horizon $h = 0, \ldots, 6$, we estimate the signed local projection (Jordà, 2005)

$$g_{j,t+h} = a^s_{jh} + \tau^s_{th} + b^{s,+}_{jh} x^{s,+}_t + b^{s,-}_{jh} x^{s,-}_t + \sum_{\ell=1}^{2} \left(c^s_{jh\ell} g_{j,t-\ell} + \boldsymbol{d}^{s\prime}_{jh\ell} \boldsymbol{C}^s_{t-\ell} \right) + u^s_{j,t+h}, \tag{24}$$

[29] For earlier evidence on how responses vary with firm size and age, see Gertler and Gilchrist (1994); Fort et al. (2013).

where $g_{jt} = \log S_{jt} - \log S_{j,t-4}$ is year-on-year real-sales growth, $x_t^{s,+} = \max\{x_t^s, 0\}$, and $x_t^{s,-} = \min\{x_t^s, 0\}$. The vector $\boldsymbol{C}^s$ contains log real GDP, the log GDP deflator, and the excess bond premium in levels. We include two lags of own growth and each macro variable, allowing all coefficients, including intercepts, to vary by spell. This accommodates heterogeneity in both growth persistence and exposure to aggregate conditions. Time fixed effects τ_{th}^s absorb common movements. The coefficients $b_{jh}^{s,+}$ therefore capture responses to growth-contracting innovations relative to other firms. We study the variance and skewness of these estimated responses.

Distributional comparison. For each shock, we select the horizon $h_s^\star$ at which the unweighted mean adverse response is most negative. We compute this mean across all estimated spells using an auxiliary regression with the same regressors and observations but without time fixed effects. This focuses the comparison on the strongest average contraction. As responses dissipate, estimation error can account for an increasing share of their estimated dispersion.

At $h_s^\star$, we retain all adverse coefficients and compute their variance and standardized skewness separately for small and large firms within industry–date cells at adverse-shock dates. We take small-minus-large differences, then average with equal weights across industries within each date and across dates.[30] These differences summarize the size gradients in dispersion and asymmetry: a positive variance difference indicates more dispersed responses among small firms, while a positive skewness difference indicates more negatively skewed responses among large firms. The model implies both differences should be positive.

Coefficient uncertainty and inference. Firm-level shock responses are estimated with error, so dispersion in their estimated coefficients reflects both true response heterogeneity and estimation uncertainty. If small firms' responses are estimated less precisely, this error could contribute to their greater measured dispersion. We therefore adjust coefficient variance by subtracting the estimated contribution of coefficient noise, accounting for covariance between estimation errors across firms.

For a cell of n firms, write the estimated coefficient vector as $\widehat{\boldsymbol{b}} = \boldsymbol{b} + \boldsymbol{e}$. Let $M_n = I_n - \mathbf{1}_n\mathbf{1}_n'/n$ be the centering matrix and $\widehat{\Omega}_e$ the estimated covariance matrix

[30] Eligible cells contain at least ten stable-small and ten stable-large firms, and eligible dates contain at least five qualifying industries. Cells and aggregation weights remain fixed across bootstrap draws. Appendix E.3 provides the complete construction.

of coefficient errors. The adjusted variance is

$$\widehat{V}^{\text{adj}} = \frac{1}{n}\widehat{\boldsymbol{b}}' M_n \widehat{\boldsymbol{b}} - \frac{1}{n}\operatorname{tr}(M_n \widehat{\Omega}_e). \tag{25}$$

The first term is observed coefficient variance; the second is its estimated noise component. Appendix E.3 derives the correction, and Appendix E.5 reports the unadjusted comparison.

Standardized skewness requires a different treatment: coefficient uncertainty matters through both the third moment and the variance used to standardize it. Under centered, symmetric coefficient error independent of the true response, the third moment is unchanged while variance increases. Observed skewness therefore equals $r^{3/2}\Gamma$, where Γ is true response skewness and $r = V(b)/[V(b) + V(e)]$ is coefficient reliability. Comparable reliability across small and large firms thus implies comparable attenuation of their skewness.

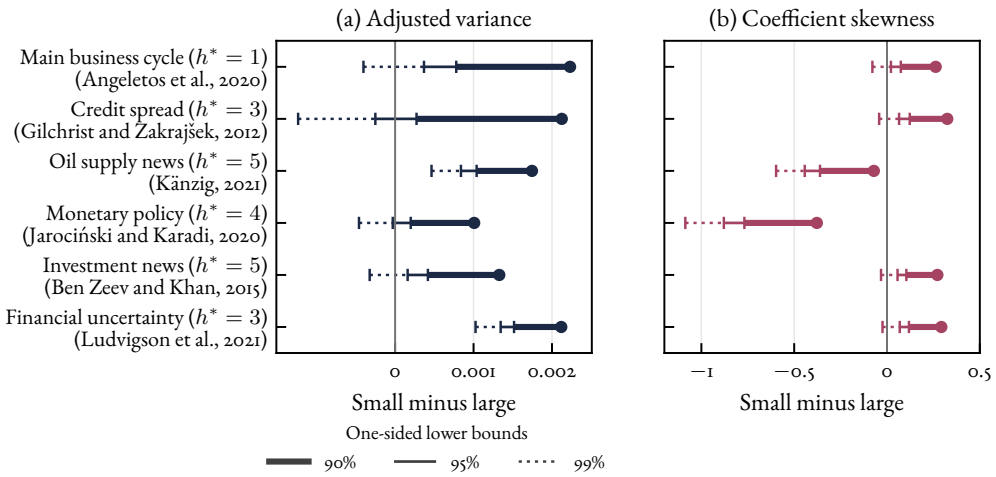
We report the skewness of the estimated response coefficients without adjustment. Table E.2 shows nearly identical average attenuation factors across the six shocks: 0.227 for small firms and 0.234 for large firms. The small-firm factor is lower for three shocks and higher for the other three, indicating no systematic size ordering in attenuation. Correcting by a common, fixed attenuation factor of about 0.23 would multiply the point estimates and confidence bounds in Figure 6(b) by approximately 4.3, leaving their signs and statistical conclusions unchanged.

We estimate $\widehat{\Omega}_e$ and conduct inference with 1,000 six-quarter calendar-block wild bootstrap draws using OLS residuals. Block multipliers are shared across firms and shocks, with reporting cells fixed across draws. We report one-sided 90, 95 and 99 percent pointwise lower confidence bounds based on the bootstrap standard error (Appendix E.4).

Results. Figure 6 provides the clearest support for the model's joint ordering under the main business-cycle, investment-news, and uncertainty shocks. For these three shocks, both adjusted-variance and skewness contrasts are positive, with 95 percent lower bounds above zero. The same holds for the adjusted-variance contrast under oil shocks and the skewness contrast under credit shocks. Skewness contrasts are 0.26, 0.32, 0.27, and 0.29 for the main business-cycle, credit, investment-news, and uncertainty shocks, respectively. The remaining 95 percent lower bounds fall below zero.

The dispersion ordering is also present before the variance correction: all six shocks have positive unadjusted variance contrasts, with 95 percent lower

Figure 6: Shock-Mediated Distributional Responses by Firm Size



Note: Small-minus-large coefficient moments at the mean-selected horizon $h_s^\star$. Moments are formed within two-digit NAICS industry–date cells and averaged equally across industries within dates and across dates. Variance is adjusted according to Equation (25); skewness is unadjusted. Bars show one-sided 90, 95 and 99 percent pointwise lower bounds from 1,000 six-quarter block-wild draws; thick, thin and dotted segments indicate the respective confidence levels.

bounds above zero. The correction therefore does not create this ordering. The HAC(8) correction likewise yields positive 95 percent lower bounds for all six shocks (Appendix E.5).

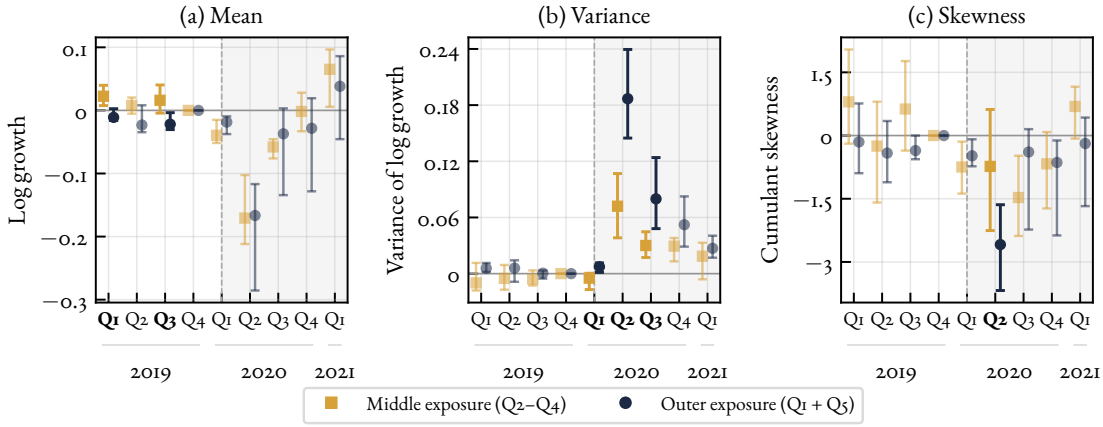
5.4 Measured Exposure and Distributional Responses

Proposition 6 assigns heterogeneous exposures a central role in the transmission of aggregate shocks. Holding policy shape and idiosyncratic risk fixed, greater exposure dispersion raises the aggregate contribution to growth variance. Policy curvature also turns this dispersed incidence into asymmetric growth. We now ask whether groups with more dispersed exposures exhibit larger distributional changes during an aggregate contraction, in support of our mechanism.

We study this question using the COVID exposures of Davis et al. (2025).[31] Their measure combines firms' pre-pandemic risk disclosures with equity returns on COVID news days. We compare the outer quintiles (1 and 5) of these signed scores with the middle quintiles (2–4) around the COVID contraction. Pooling both tails targets dispersion in exposure: the outer group contains firms at both extremes of the measured response, whereas the middle group has more similar

[31] Related work exploits differential exposure to aggregate energy, currency, and policy risks to identify uncertainty effects (Alfaro et al., 2024).

Figure 7: COVID Exposure and Growth-Distribution Changes



Note: Changes from 2019Q4 at common industry–size shares. Yellow squares: exposure quintiles 2–4; blue circles: 1 and 5. Bars: 90% pointwise percentile intervals from 5,000 industry-pairs bootstrap draws. Bold labels and darker markers and whiskers denote significant outer-minus-middle changes.

exposures. To follow the same firms through the COVID contraction and recovery, we require observed annual real-sales growth in every quarter from 2019Q1 to 2021Q1. Exposure groups remain fixed over this window. Within each two-digit NAICS industry, we classify firms as small or large using median pre-event size. The minimum cell-size requirement leaves 906 firms across eight industries.[32]

The two exposure groups may nevertheless differ in industry and size composition. Because firm size is linked to policy shape in our framework, these differences can themselves generate different distributional responses. Following Chernozhukov et al. (2013), we therefore give each industry–size cell the same weight in both groups, equal to its share in the combined sample. We then compute mean growth, variance, and skewness from each group's reweighted firm-growth distribution, retaining variation both within and across cells.

Figure 7 plots each group's moment changes relative to 2019Q4, removing baseline differences in moment levels. Bold quarter labels indicate statistically significant differences between the groups' moment changes. In 2020Q2, mean growth falls by about 0.17 log units in both groups. Their higher moments respond differently: variance rises by 0.187 among outer-exposure firms and 0.072 among middle-exposure firms, while skewness falls by 2.59 and 0.73, respectively. The larger variance increase and larger skewness decline in the outer group are both

[32]Pre-event size is the prior-four-quarter average of real sales measured at 2019Q4. Each industry–size cell must contain at least eight firms from each exposure group.

statistically significant at the 10% level under pointwise inference.[33] The larger rise in variance and fall in skewness among firms with more dispersed exposures are consistent with the mechanism's qualitative predictions: heterogeneous exposure spreads the incidence of the contraction, while concave policies make growth more negatively skewed.

We repeat the exercise with inflation exposures in Appendix D.4. In that event setting, the skewness contrast has the opposite sign. However, unlike COVID, the inflation episode lacks a sharply dated onset and overlaps the pandemic recovery and other aggregate news. The exposure groups also display substantial pre-event differences in variance and skewness. These features make the inflation episode less informative about the response to a common shock. We therefore place greater weight on the COVID results, which provide clear support for our exposure mechanism.

6 Conclusion

This paper shows how heterogeneous exposure and demand curvature allow a common aggregate shock to jointly move mean growth, dispersion, and skewness in a general homothetic framework of monopolistic competition. The framework also explains why larger firms can have lower and more stable growth dispersion yet larger skewness fluctuations. Evidence from U.S. public firms supports the predicted differences in shock transmission. The estimated slopes and curvature magnitudes also provide quantitative targets for models without a demand microfoundation.

These results show how business-cycle moment dynamics can inform the study of demand. Within our framework, the joint behavior of dispersion and skewness provides aggregate evidence for Marshall's Second Law and its role in firms' responses to shocks. Summarily, our analysis reveals a deep relationship between distributional business cycles, the transmission of shocks and price-setting power.

The methods developed in this paper offer a parsimonious foundation for understanding the joint behavior of business-cycle moments. We believe that a promising direction for future research is to study the aggregate sources of firm-level shocks and the heterogeneity of firms' exposures to them (recent advances were made by Davis et al., 2025), and to examine the reactions to aggregate shock with firm-level micro data, which could lead to a further tightening of the link between market power and aggregate fluctuations, or establish additional explanations for

[33]The bootstrap resamples among industries, preserving firm histories and exposure-group masses and recomputing weights and moments.

distributional business cycles.

Appendix

Model Proofs

Proof of Proposition 1. Write $s = s(u)$ and substitute $q_i = sx_i$ into the firm's problem, holding conjectured rival output at $s\bar{\boldsymbol{q}}_{-i}$. Since h is homogeneous of degree one and h_i of degree zero,

$$R_i(sx; s\bar{\boldsymbol{q}}_{-i}) = E\frac{sx\, h_i(sx, s\bar{\boldsymbol{q}}_{-i})}{h(sx, s\bar{\boldsymbol{q}}_{-i})} = E\frac{x\, h_i(x, \bar{\boldsymbol{q}}_{-i})}{h(x, \bar{\boldsymbol{q}}_{-i})} = R_i(x; \bar{\boldsymbol{q}}_{-i}).$$

The cost term becomes $\phi_i e^{\bar{\lambda}u+\epsilon_i}(s(u)x)^\eta = \phi_i e^{\epsilon_i} x^\eta$. The transformed problem is therefore the zero-aggregate-shock problem defining $x_i(\epsilon_i)$. Uniqueness gives $q_i = s(u)x_i(\epsilon_i)$, and $x_i(0) = \bar{q}_i$.

At actual output $\boldsymbol{q} = s(u)\boldsymbol{x}(\boldsymbol{\epsilon})$, homogeneity of inverse demand and the price index gives the prices stated above. Euler's theorem implies $\int_I \widetilde{p}_i q_i\, di = \frac{E}{h(\boldsymbol{q})}\int_I h_i(\boldsymbol{q}) q_i\, di = E$. Concavity of h establishes household optimality, so these prices clear markets.

Deflated sales are $r_i(u, \boldsymbol{\epsilon}) = s(u)x_i(\epsilon_i)h_i(\boldsymbol{x}(\boldsymbol{\epsilon}))$. Taking logs and adding and subtracting $\log h_i(x_i(\epsilon_i), \bar{\boldsymbol{q}}_{-i})$ gives the real-revenue representation in Equation (5). The output representation follows directly from $q_i = s(u)x_i(\epsilon_i)$. ■

Firm Problem and Regularity

Assumption 1 (Regularity Conditions)**.** *The inverse demand function p satisfies:*

i) $p : \mathbb{R}_+ \to \mathbb{R}_+$, $p \in \mathcal{C}^3(\mathbb{R}_+)$, and $p'(q) < 0$ on the relevant operating domain,

ii) marginal revenue $R'(q) := p(q) + qp'(q)$ is strictly decreasing on the relevant operating domain and positive at some finite quantity level.

The condition on marginal revenue is implied by log-concavity of inverse demand. If p is log-concave and downward sloping, then marginal revenue is strictly decreasing wherever it is positive. It is not necessary: CES inverse demand satisfies the condition as well, although it is not log-concave. The maintained restriction $\eta > 1$ also rules out degenerate zero-or-infinity solutions for the price-taking benchmark.

Lemma 1 (Solution of Firm Problem)**.** *Under Assumption 1, the firm's solution is unique, interior, and characterized by Equations (7) and (8). Optimal log output,*

$\mathcal{Q}(\epsilon)$, *is decreasing in* ϵ. *On the relevant domain, inverse-demand elasticity satisfies* $\mathcal{E}p(q) \in (-1, 0)$ *and the markup is* $\mu(q) = \big(1 + \mathcal{E}p(q)\big)^{-1}$.

For the proofs below, write $\widehat{q} := \log q$.

Proof of Lemma 1. We first compare marginal revenue with marginal cost and then invert the log first-order condition. Differentiating profits gives $\pi'(q) = R'(q) - \eta\phi e^{\epsilon} q^{\eta-1}$. Marginal revenue is strictly decreasing and marginal cost strictly increasing, so the maintained interior crossing is unique and satisfies Equation (7). Taking logs gives $\epsilon = H(\widehat{q})$, with

$$H'(\widehat{q}) = \frac{e^{\widehat{q}} R''(e^{\widehat{q}})}{R'(e^{\widehat{q}})} - (\eta - 1) = m(e^{\widehat{q}}) - (\eta - 1) < 0.$$

Hence the inverse policy $\mathcal{Q} = H^{-1}$ is decreasing. Finally, $R'(q) = p(q)[1 + \mathcal{E}p(q)] > 0$ and $p'(q) < 0$ imply $\mathcal{E}p(q) \in (-1, 0)$, while dividing price by marginal cost gives $\mu(q) = [1 + \mathcal{E}p(q)]^{-1}$. ■

Lemma 2 (Local Curvature of Output)**.** *Let* $\widehat{q}^\star := \mathcal{Q}(\epsilon^\star)$ *and* $q^\star := e^{\widehat{q}^\star}$. *If* $H'(\widehat{q}^\star) \neq 0$, *then*

$$\mathcal{Q}'(\epsilon^\star) = \frac{1}{H'(\widehat{q}^\star)}, \qquad \mathcal{Q}''(\epsilon^\star) = -\frac{H''(\widehat{q}^\star)}{H'(\widehat{q}^\star)^3}, \qquad \rho^\star_{\mathcal{Q}} = \frac{H''(\widehat{q}^\star)}{H'(\widehat{q}^\star)^2}, \tag{26}$$

$$H'(\widehat{q}^\star) = m(q^\star) - (\eta - 1), \qquad H''(\widehat{q}^\star) = \left.\frac{dm(q)}{d\log q}\right|_{q=q^\star}. \tag{27}$$

Thus $m'(q^\star) < 0$ *is equivalent to* $\mathcal{Q}''(\epsilon^\star) < 0$ *and* $\rho^\star_{\mathcal{Q}} < 0$. *In particular, MSLD* implies local concavity, whereas anti-MSLD* implies local convexity.*

Proof of Lemma 2. Our argument differentiates the inverse identity $H(\mathcal{Q}(\epsilon)) = \epsilon$. Once and twice differentiation give

$$H'(\mathcal{Q})\mathcal{Q}' = 1, \qquad H''(\mathcal{Q})(\mathcal{Q}')^2 + H'(\mathcal{Q})\mathcal{Q}'' = 0,$$

and hence $\mathcal{Q}' = 1/H'$ and $\mathcal{Q}'' = -H''/H'^3$. At the optimum, $m(q^\star) < 0$ and $\eta > 1$, so $H'(\widehat{q}^\star) < 0$ and $|\mathcal{Q}'| = -1/H'$. Therefore $\rho^\star_{\mathcal{Q}} = H''/H'^2$.

Differentiating the definition of H yields

$$H'(\widehat{q}) = \frac{e^{\widehat{q}} R''(e^{\widehat{q}})}{R'(e^{\widehat{q}})} - (\eta - 1) = m(e^{\widehat{q}}) - (\eta - 1), \qquad H''(\widehat{q}) = e^{\widehat{q}} m'(e^{\widehat{q}}).$$

Thus $H''(\widehat{q}^\star) = q^\star m'(q^\star)$ has the sign of $m'(q^\star)$. The formulas for $\mathcal{Q}''$ and $\rho^\star_{\mathcal{Q}}$ give the stated equivalence and the curvature implications of MSLD* and anti-MSLD*. ■

Demand-Policy Proofs

Proof of Proposition 2. We apply the inverse-function identities stated in Lemma 2 to the marginal-revenue elasticity. Differentiating H at $\widehat{q}^\star$ gives $H'(\widehat{q}^\star) = m(q^\star) - (\eta - 1) = -D(q^\star)$ and $H''(\widehat{q}^\star) = q^\star m'(q^\star)$. Since $m(q^\star) < 0$ and $\eta > 1$, $D(q^\star) > 0$, and therefore $\mathcal{Q}'(0) = 1/H'(\widehat{q}^\star) = -1/D(q^\star)$. Moreover, $\rho^\star_{\mathcal{Q}} = H''(\widehat{q}^\star)/H'(\widehat{q}^\star)^2 = q^\star m'(q^\star)/D(q^\star)^2$. Because $q^\star > 0$ and the denominator is positive, output-policy curvature has the sign of $m'(q^\star)$. MSLD* therefore implies $\rho^\star_{\mathcal{Q}} < 0$, while anti-MSLD* implies $\rho^\star_{\mathcal{Q}} > 0$. ■

Proof of Corollary 1. Solving the price-taking policy explicitly gives $\mathcal{Q}_{\mathrm{pt}}(\epsilon) = [\log \bar{p} - \log \eta - \log \phi - \epsilon]/(\eta - 1)$, hence $\mathcal{Q}'_{\mathrm{pt}}(0) = -1/(\eta - 1)$ and $\rho^\star_{\mathrm{pt},\mathcal{Q}} = 0$. For a price setter, $m(q^\star) < 0$ implies $D(q^\star) > \eta - 1$, so Proposition 2 gives $|\mathcal{Q}'(0)| = 1/D(q^\star) < 1/(\eta - 1) = |\mathcal{Q}'_{\mathrm{pt}}(0)|$. The same proposition gives $\rho^\star_{\mathcal{Q}} = q^\star m'(q^\star)/D(q^\star)^2 < 0$ under MSLD*. ■

Proof of Proposition 3. Differentiate log real revenue as a composition of inverse demand and the output policy, then sign its two curvature components. Write $q(\epsilon) = e^{\mathcal{Q}(\epsilon)}$. Since $\mathcal{R}(\epsilon) = \mathcal{Q}(\epsilon) + \log p(q(\epsilon))$, the chain rule gives

$$\mathcal{R}'(\epsilon) = [1 + \varepsilon(q)]\mathcal{Q}'(\epsilon), \qquad \mathcal{R}''(\epsilon) = [1 + \varepsilon(q)]\mathcal{Q}''(\epsilon) + \varepsilon(q)\psi(q)[\mathcal{Q}'(\epsilon)]^2,$$

where $d\varepsilon(q(\epsilon))/d\epsilon = \varepsilon(q)\psi(q)\mathcal{Q}'(\epsilon)$. Because $1 + \varepsilon^\star > 0$, dividing the second identity at zero by $|\mathcal{R}'(0)| = (1 + \varepsilon^\star)|\mathcal{Q}'(0)|$ proves Equation (13).

Under MSLD*, $\varepsilon'(q^\star) < 0$ and $m'(q^\star) < 0$. Since $\varepsilon^\star < 0$, the first inequality gives $\psi^\star > 0$, making the relative-price contribution in Equation (13) negative. The second inequality gives $\rho^\star_{\mathcal{Q}} < 0$ by Proposition 2, so $\rho^\star_{\mathcal{R}} < 0$. Under anti-MSLD*, both derivative inequalities reverse: $\psi^\star < 0$, the relative-price contribution is positive, and $\rho^\star_{\mathcal{Q}} > 0$. Hence $\rho^\star_{\mathcal{R}} > 0$. Finally, a price taker has locally constant p, hence $\varepsilon = 0$ and $\mathcal{R} = \mathcal{Q} + \text{constant}$, giving zero curvature. ■

Proof of Proposition 4. At the boundary, $R'(0) = \bar{p}$. Differentiating $\varepsilon(q) = qp'(q)/p(q)$ and $m(q) = qR''(q)/R'(q)$ and taking right limits gives

$$2\varepsilon'_+(0) = m'_+(0) = \frac{R''_+(0)}{R'(0)} = \frac{2p'_+(0)}{\bar{p}} =: -\iota_{R'} < 0. \tag{28}$$

Hence $m(q) = -\iota_{R'} q + O(q^2)$ and $m'(q) = -\iota_{R'} + O(q)$. Set $A := \eta - 1 > 0$. The marginal cost–revenue elasticity gap is $D(q) = A + \iota_{R'} q + O(q^2)$. Proposition 2 gives $\beta_{\mathcal{Q}} = -1/D$ and $\rho_{\mathcal{Q}} = qm'/D^2$. Using

$$\frac{1}{D(q)} = \frac{1}{A} - \frac{\iota_{R'}}{A^2} q + O(q^2),$$

we obtain $\beta_{\mathcal{Q}}(q) = -A^{-1} + \iota_{R'} A^{-2} q + O(q^2)$ and $\rho_{\mathcal{Q}}(q) = -\iota_{R'} A^{-2} q + O(q^2)$, proving the expansions with $C_Q = \iota_{R'}/A^2$. Differentiability of the remainders under $p \in C^4$ gives $\beta'_{\mathcal{Q}}(q) = C_Q + O(q) > 0$ and $\rho'_{\mathcal{Q}}(q) = -C_Q + O(q) < 0$ for sufficiently small $q > 0$. In that neighborhood both $\beta_{\mathcal{Q}}$ and $\rho_{\mathcal{Q}}$ are negative, so $d|\beta_{\mathcal{Q}}|/dq = -\beta'_{\mathcal{Q}} < 0$ and $d|\rho_{\mathcal{Q}}|/dq = -\rho'_{\mathcal{Q}} > 0$. ■

Proof of Proposition 5. We combine the output-policy expansion in Proposition 4 with the local inverse-demand elasticity. Its boundary calculation gives $p'(0) = -\iota_{R'}\bar{p}/2$, and therefore

$$p(q) = \bar{p} - \frac{\iota_{R'}\bar{p}}{2} q + O(q^2), \qquad \varepsilon(q) = \frac{qp'(q)}{p(q)} = -\frac{\iota_{R'}}{2} q + O(q^2).$$

Because $\mathcal{R} = \log[p(q)q]$ composed with the output policy, $\beta_{\mathcal{R}}(q) = [1 + \varepsilon(q)]\beta_{\mathcal{Q}}(q)$. Set $A := \eta - 1$. Proposition 4 then gives

$$\beta_{\mathcal{R}}(q) = \left[1 - \frac{\iota_{R'}}{2} q + O(q^2)\right]\left[-\frac{1}{A} + \frac{\iota_{R'}}{A^2} q + O(q^2)\right] = -\frac{1}{\eta - 1} + \frac{\iota_{R'}}{2} \frac{\eta + 1}{(\eta - 1)^2} q + O(q^2).$$

For curvature, Proposition 3 gives $\rho_{\mathcal{R}} = \rho_{\mathcal{Q}} + \varepsilon\psi|\beta_{\mathcal{Q}}|/(1 + \varepsilon)$. Writing $\ell := \iota_{R'}/2$, the expansion $\varepsilon(q) = -\ell q + O(q^2)$ implies $\psi(q) = q\varepsilon'(q)/\varepsilon(q) = 1 + O(q)$. Also $|\beta_{\mathcal{Q}}| = A^{-1} + O(q)$ and $1 + \varepsilon = 1 + O(q)$. Hence

$$\rho_{\mathcal{R}}(q) = -\frac{\iota_{R'}}{A^2} q - \frac{\iota_{R'}}{2A} q + O(q^2) = -\frac{\iota_{R'}}{2} \frac{\eta + 1}{(\eta - 1)^2} q + O(q^2).$$

Differentiating gives the stated local signs, exactly as in Proposition 4. Finally, $d[p(q)q]/dq = R'(q) > 0$ locally, so revenue preserves the quantity ranking. ■

Propositions 4 and 5 use a finite, nonzero slope at the choke price. The following extension permits inverse demand to be flat or infinitely steep there.

Remark 2 (A regular choke price)**.** *Suppose that, for some $c, \gamma > 0$, inverse demand and the derivatives entering the policy formulas satisfy $p(q) = \bar{p} - cq^\gamma + o(q^\gamma)$ as $q \downarrow 0$. With $L(q) := -\mathcal{E}p(q)$, one then has*

$$\begin{aligned}
L(q) &\sim \frac{c\gamma}{\bar{p}} q^\gamma, & m(q) &= -(1 + \gamma)L(q) + o(L(q)), \\
\beta_{\mathcal{Q}}(q) &= -\frac{1}{\eta - 1} + \frac{1 + \gamma}{(\eta - 1)^2} L(q) + o(L(q)), & \rho_{\mathcal{Q}}(q) &= -\frac{\gamma(1 + \gamma)}{(\eta - 1)^2} L(q) + o(L(q)), \\
\beta_{\mathcal{R}}(q) &= -\frac{1}{\eta - 1} + \frac{\eta + \gamma}{(\eta - 1)^2} L(q) + o(L(q)), & \rho_{\mathcal{R}}(q) &= -\frac{\gamma(\eta + \gamma)}{(\eta - 1)^2} L(q) + o(L(q)).
\end{aligned}$$

Hence the same local slope and curvature gradients obtain. The Klenow–Willis Kimball inverse demand (Klenow and Willis, 2016) satisfies this regular-choke-price condition for every positive Kimball curvature parameter, with regular-choke exponent $\gamma = \nu$.

In this expansion, $\gamma = 1$ gives a finite, nonzero boundary slope, $\gamma > 1$ allows inverse demand to be flat, and $0 < \gamma < 1$ allows it to be infinitely steep. Each case delivers the same local decline in absolute policy slopes and increase in normalized curvature magnitude with firm size.

Proposition 8 (CES as a knife-edge benchmark)**.** *If $p(q) = Aq^{-\chi}$ with $A > 0$ and $\chi \in (0, 1)$, then $\varepsilon(q) \equiv m(q) \equiv -\chi$ and $\psi(q) \equiv 0$. Consequently, $\rho^\star_{\mathcal{Q}} = \rho^\star_{\mathcal{R}} = 0$ at every operating quantity. CES therefore remains the knife-edge case in which the output policy is locally linear, the direct real-revenue-curvature term vanishes, and skewness is absent to second order.*

Proof of Proposition 8. Under CES, both the marginal-revenue elasticity and the log first-order condition are affine. Differentiating $p(q) = Aq^{-\chi}$ gives $\varepsilon(q) = qp'(q)/p(q) = -\chi$, hence $\psi(q) = 0$. Moreover,

$$R'(q) = p(q) + qp'(q) = A(1-\chi)q^{-\chi}, \qquad m(q) = \frac{qR''(q)}{R'(q)} = -\chi.$$

The log first-order condition is therefore

$$\epsilon = \log A + \log(1-\chi) - \log\eta - \log\phi - (\eta - 1 + \chi)\widehat{q},$$

so its inverse $\mathcal{Q}(\epsilon)$ is affine and $\rho^\star_{\mathcal{Q}} = 0$. Equation (13) then gives $\rho^\star_{\mathcal{R}} = 0$ because both $\rho^\star_{\mathcal{Q}}$ and ψ vanish. Proposition 6's leading skewness term is proportional to the relevant curvature index and therefore vanishes for output and real sales. ■

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Supplemental Online Appendix

Monopolistically Skewed Business Cycles: On the Origins of Distributional Fluctuations

A Price-Residual Bounds

Proposition A.1 (Price Residual Condition in Symmetric Equilibrium)**.** *Consider the equilibrium in Proposition 1 with symmetric demand, equal baseline cost shifters, and the shock structure of Section 3.1. The price residual admits the decomposition*

$$\zeta_i(\boldsymbol{\epsilon}) = Z(\boldsymbol{\epsilon}) + \nu_i(\boldsymbol{\epsilon}), \qquad \|\nu\|_{L^3} := \left(\mathbb{E}_i|\nu_i|^3\right)^{1/3} = O(\sigma_\epsilon^3). \tag{A.1}$$

More strongly, $\nu_i = O(\sigma_\epsilon^3)$ uniformly in i.

Proof of Proposition A.1. We express the price-forecast error as a smooth function of own and rival log-output deviations. A second-order expansion in rivals' deviations depends only on their cross-sectional mean and variance; symmetry makes both common across firms at second order, while interactions with the firm's own first-order deviation are third order.

Fix t and define the shock scale $s := \sigma_{\epsilon,t} = (\sigma_\kappa^2 + u_t^2\sigma_\lambda^2)^{1/2}$. All order statements below refer to $s \downarrow 0$. Symmetry gives a common benchmark $\bar{q}$, policy $x(\epsilon)$, and log policy $\mathcal{Q}(\epsilon) = \log x(\epsilon)$. Define $\delta_{it} := \log x(\epsilon_{it}) - \log \bar{q}$. Bounded standardized shocks and smoothness imply $\epsilon_{it} = O(s)$ and $\delta_{it} = O(s)$ uniformly. Taylor expansion gives

$$\delta(e) = \mathcal{Q}'(0)e + \frac{1}{2}\mathcal{Q}''(0)e^2 + O(e^3).$$

Since $\mathbb{E}e = 0$, $\mathbb{E}e^2 = \sigma_{\epsilon,t}^2$, and $\mathbb{E}|e|^3 = O(s^3)$, the cross-sectional mean and variance of rival deviations satisfy

$$M_t := \mathbb{E}_i\delta_{it} = \frac{1}{2}\mathcal{Q}''(0)\sigma_{\epsilon,t}^2 + O(s^3) = O(s^2),$$
$$V_t := \mathrm{Var}_i(\delta_{it}) = \mathcal{Q}'(0)^2\sigma_{\epsilon,t}^2 + O(s^3) = O(s^2).$$

To make the price-forecast error explicit, approximate the continuum by N symmetric rivals. For rival deviations z_j, define

$$G_N(\delta, z_1, \ldots, z_N) := \log h_i\left(\bar{q}e^{\delta}, \{\bar{q}e^{z_j}\}_{j=1}^N\right).$$

The first argument is $\bar{q}e^{\delta}$ because $\delta = \log q_i - \log \bar{q}$. The exact finite-$N$ forecast error is

$$\zeta_{i,N} = G_N(\delta_{it}, \boldsymbol{z}_t) - G_N(\delta_{it}, \mathbf{0}) :$$

the first term evaluates the realized relative-price derivative at actual rival quantities,

while the second holds every rival at $\bar{q}$. At $\boldsymbol{z} = \mathbf{0}$, a second-order Taylor expansion gives

$$G_N(\delta, \boldsymbol{z}) - G_N(\delta, \mathbf{0}) = \sum_j G_j(\delta) z_j + \frac{1}{2} \sum_j \sum_k G_{jk}(\delta) z_j z_k + O(\|\boldsymbol{z}\|^3).$$

Symmetry across rivals implies $G_j = a_N$, $G_{jj} = b_N$, and $G_{jk} = c_N$ for $j \neq k$, with coefficients depending on δ. Let $\bar{z}_N = N^{-1} \sum_j z_j$ and $\overline{z^2}_N = N^{-1} \sum_j z_j^2$. Because

$$\sum_{j \neq k} z_j z_k = N^2 \bar{z}_N^2 - N \overline{z^2}_N,$$

the expansion becomes

$$G_N(\delta, \boldsymbol{z}) - G_N(\delta, \mathbf{0}) = A_N(\delta) \bar{z}_N + B_N(\delta) \overline{z^2}_N + C_N(\delta) \bar{z}_N^2 + O(\|\boldsymbol{z}\|^3),$$

where $A_N = N a_N$, $B_N = N(b_N - c_N)/2$, and $C_N = N^2 c_N / 2$.

Assuming these coefficient combinations and the third-order remainder have finite uniform local continuum limits, the law of large numbers gives $\bar{z}_N \to M_t$ and $\overline{z^2}_N \to V_t + M_t^2$. Since $M_t^2 = O(s^4)$,

$$\zeta_{it} = A(\delta_{it}) M_t + B(\delta_{it}) V_t + O(s^3).$$

Smoothness in own deviation implies $A(\delta_{it}) = A_0 + A_1 \delta_{it} + O(\delta_{it}^2)$ and analogously for B. Because $\delta_{it} = O(s)$ while $M_t, V_t = O(s^2)$, every term involving $\delta_{it} M_t$ or $\delta_{it} V_t$ is third order. Therefore

$$\zeta_{it} = \left[\frac{1}{2} A_0 \mathcal{Q}''(0) + B_0 \mathcal{Q}'(0)^2 \right] \sigma_{\epsilon,t}^2 + O(s^3).$$

Define $\chi_h := A_0 \mathcal{Q}''(0)/2 + B_0 \mathcal{Q}'(0)^2$ and $Z_t := \chi_h(\sigma_\kappa^2 + u_t^2 \sigma_\lambda^2)$. Then $\zeta_{it} = Z_t + \nu_{it}$ with $\sup_i |\nu_{it}| = O(s^3)$, so $\|\nu_t\|_{L^3(i)} \leq \sup_i |\nu_{it}| = O(s^3)$. Also $\Delta Z_t = \chi_h \sigma_\lambda^2 \Delta(u_t^2)$. ■

Under symmetry, the common price residual can affect mean log real sales, while differences across firms enter only at third order. We next allow baseline productivity and demand to differ across public types.

Remark A.1 (Breaking symmetry)**.** *The analysis below extends the result to finitely many positive-mass public types $\tau = (\phi_\tau, p_\tau)$, where ϕ_τ is the baseline cost shifter and p_τ is the residual inverse-demand schedule, including its dependence on competitors' quantities. These characteristics may differ across types, while firms are symmetric within each type.*

Homogeneity continues to give the exact output representation $q_{i\tau}(u, \epsilon_{i\tau}) = s(u) x_\tau(\epsilon_{i\tau})$. Under the maintained shock conditions, the price residual satisfies $\zeta_{i\tau} = Z_\tau + \nu_{i\tau}, \quad \|\nu_\tau\|_{L^3} = O(s^3)$, where s is the largest type-specific shock scale. The common component Z_τ may differ across types. Within each type, it contributes only a common shift to log real sales, leaving the leading variance and skewness approximations in Proposition 6 unchanged. With a common inverse-demand

schedule, groups may be defined by baseline productivity through ϕ_τ.

Finite Public Types

Partition the unit mass of firms into positive-mass sets I_τ, and let τ collect a type's baseline cost and residual-demand environment. Assume demand is type-anonymous—invariant to relabeling firms within I_τ—and impose the symmetry, independence, bounded-support, and moment conditions of Section 3.1 conditional on type. In particular, shocks are centered within type. By homogeneity, every type has its own policy and the same aggregate scale:

$$q_{i\tau}(u, \epsilon_{i\tau}) = e^{-\bar{\lambda}u/\eta} x_\tau(\epsilon_{i\tau}), \qquad \log r_{i\tau,t} = -\frac{\bar{\lambda}}{\eta} u_t + \mathcal{R}_\tau(\epsilon_{i\tau,t}) + \zeta_{i\tau,t}. \tag{A.2}$$

For each receiving type τ, impose the type-anonymous version of the second-order functional expansion above. Rivals' quantities enter through the first two log-deviation moments of every source type. The expansion and its cubic remainder hold uniformly over the finite type set.

Writing $\delta_{i\tau,t}$ for own log-quantity deviation and $M_{\sigma t}, S_{\sigma t}$ for those two source moments, use the coefficient convention $\zeta_{i\tau,t} = \sum_\sigma \{a_{\tau\sigma}(\delta_{i\tau,t}) M_{\sigma t} + \frac{1}{2} b_{\tau\sigma}(\delta_{i\tau,t}) S_{\sigma t}\} + O(s_t^3)$. Let $\sigma^2_{\epsilon,\sigma,t} = \sigma^2_{\kappa,\sigma} + u_t^2 \sigma^2_{\lambda,\sigma}$ and $s_t := \max_\sigma \sigma_{\epsilon,\sigma,t}$. Conditional centering and a Taylor expansion of $\mathcal{Q}_\sigma$ imply that the type-σ first and second raw log-quantity moments are, respectively, $\frac{1}{2}\mathcal{Q}''_\sigma(0)\sigma^2_{\epsilon,\sigma,t} + O(s_t^3)$ and $\mathcal{Q}'_\sigma(0)^2 \sigma^2_{\epsilon,\sigma,t} + O(s_t^3)$. Substitution in the functional expansion gives

$$\begin{aligned} \zeta_{i\tau,t} &= Z_{\tau t} + \nu_{i\tau,t}, & \|\nu_{\tau t}\|_{L^3(I_\tau)} &= O(s_t^3), \\ Z_{\tau t} &= \sum_\sigma \chi_{\tau\sigma} \sigma^2_{\epsilon,\sigma,t}, & \chi_{\tau\sigma} &:= \frac{1}{2}\left[a_{\tau\sigma}\mathcal{Q}''_\sigma(0) + b_{\tau\sigma}\mathcal{Q}'_\sigma(0)^2\right], \end{aligned} \tag{A.3}$$

where $a_{\tau\sigma}$ and $b_{\tau\sigma}$ are the corresponding type-moment derivatives at the benchmark. With $s := \max\{s_t, s_{t-1}\}$,

$$\mathrm{Var}_\tau(g_t) = \mathrm{Var}_\tau(\Delta\mathcal{R}_\tau(\epsilon_t)) + O(s^4), \qquad \mu_{3,\tau}(g_t) = \mu_{3,\tau}(\Delta\mathcal{R}_\tau(\epsilon_t)) + O(s^5).$$

If types share the same inverse-demand curve and shock process but differ in baseline costs, the local operating-quantity comparison in Propositions 4 and 5 also ranks their policy slopes and curvature near the choke quantity. Pooling types introduces the standard between-type mean terms, so an arbitrary empirical bin requires the group-level residual condition in Equation (A.1) rather than automatic exact cancellation.

B Growth Moments and Identification

B.1 Growth-Moment Proofs

Proof of Proposition 6. We proceed in three steps: expand the fixed policy to second order, calculate the first three cross-sectional moments, and factor the policy coefficients out of the time-series moments. Fix t, condition on $\mathcal{F}_t$, and set

$$X_1 := \epsilon_{it} = \kappa_{it} + u_t\lambda_i, \;\; X_0 := \epsilon_{i,t-1} = \kappa_{i,t-1} + u_{t-1}\lambda_i, \;\; D := X_1 - X_0, \;\; H := X_1^2 - X_0^2.$$

Under the maintained local scaling and moment bounds, $D = O(s_t)$ and $H = O(s_t^2)$ in the norms used below. Consequently, $\mathbb{E}H = O(s_t^2)$, $\mathrm{Var}(D) = \Theta(s_t^2)$ whenever the conditional variance is nondegenerate, and $\mathrm{Cov}(D^2, H) = O(s_t^4)$. Writing $r_1 := \mathcal{R}'(0)$ and $r_2 := \mathcal{R}''(0)$, a second-order expansion gives $\mathcal{R}(X_1) - \mathcal{R}(X_0) = r_1 D + r_2 H/2 + O(s_t^3)$. The common component $-b_R\Delta u_t + \Delta Z_t$ changes only the mean, while $\Delta\nu_{it} = O_{L^3}(s_t^3)$ contributes beyond the retained orders.

Because $\mathbb{E}\kappa_{it} = \mathbb{E}\lambda_i = 0$, $\mathbb{E}D = 0$, whereas

$$\mathbb{E}H = (\sigma_\kappa^2 + u_t^2\sigma_\lambda^2) - (\sigma_\kappa^2 + u_{t-1}^2\sigma_\lambda^2) = \Delta(u_t^2)\sigma_\lambda^2 = O(s_t^2),$$

which proves Equation (17). Moreover,

$$\mathrm{Var}(D) = \mathrm{Var}(\kappa_{it} - \kappa_{i,t-1}) + (\Delta u_t)^2\mathrm{Var}(\lambda_i) = 2\sigma_\kappa^2 + (\Delta u_t)^2\sigma_\lambda^2 = \Theta(s_t^2).$$

Only the linear policy term contributes at second order, so $\mathrm{Var}(r_1 D + r_2 H/2) = r_1^2\mathrm{Var}(D) + O(s_t^3)$, proving Equation (18).

For the leading third central moment, let $H_0 := H - \mathbb{E}H$. Centering and cubing yields

$$\mathbb{E}\left(r_1 D + \frac{r_2}{2}H_0\right)^3 = r_1^3\mathbb{E}D^3 + \frac{3}{2}r_1^2 r_2\mathbb{E}[D^2 H_0] + O(s_t^5).$$

The joint distribution of $(\kappa_{it}, \kappa_{i,t-1}, \lambda_i)$ is invariant under multiplication by -1, so D is symmetric and $\mathbb{E}D^3 = 0$. Thus

$$\mu_{3,t} = \frac{3}{2}r_1^2 r_2\mathrm{Cov}(D^2, H) + O(s_t^5). \tag{B.1}$$

To evaluate the covariance, abbreviate $d := u_t - u_{t-1}$, $s := u_t + u_{t-1}$, $A := \kappa_{it} - \kappa_{i,t-1}$, $K := \kappa_{it}^2 - \kappa_{i,t-1}^2$, and $J := u_t\kappa_{it} - u_{t-1}\kappa_{i,t-1}$. Then $D = A + d\lambda_i$ and $H = K + 2\lambda_i J + sd\lambda_i^2$. Expanding $D^2 = A^2 + 2dA\lambda_i + d^2\lambda_i^2$, independence and symmetry eliminate every pairing whose expectation factors through an odd moment of a centered symmetric variable. For example, $\mathrm{Cov}(A^2, 2\lambda_i J) = 2\mathbb{E}[\lambda_i]\mathbb{E}[A^2 J] = 0$, while $\mathrm{Cov}(A^2, sd\lambda_i^2) = 0$ by independence. The surviving terms are

$$\mathrm{Cov}(D^2, H) = \mathrm{Cov}(A^2, K) + \mathrm{Cov}(2dA\lambda_i, 2\lambda_i J) + \mathrm{Cov}(d^2\lambda_i^2, sd\lambda_i^2).$$

The first is zero because the two iid idiosyncratic shocks enter symmetrically. Since

$\mathbb{E}[AJ] = (u_t + u_{t-1})\sigma_\kappa^2 = s\sigma_\kappa^2$, the second is $4ds\sigma_\kappa^2\sigma_\lambda^2$; the third is $sd^3\mathrm{Var}(\lambda^2)$. Hence

$$\mathrm{Cov}(D^2, H) = 4(\Delta u_t)(u_t + u_{t-1})\sigma_\kappa^2\sigma_\lambda^2 + \Delta(u_t^2)(\Delta u_t)^2\mathrm{Var}(\lambda^2) = O(s_t^4).$$

Let $V_{D,t} := 2\sigma_\kappa^2 + d^2\sigma_\lambda^2 = \Theta(s_t^2)$. The variance expansion implies

$$v_t^{3/2} = |r_1|^3 V_{D,t}^{3/2} + O(s_t^4).$$

Since $\mathrm{Cov}(D^2, H) = O(s_t^4)$, dividing Equation (B.1) by $v_t^{3/2}$ and using $r_2/|r_1| = \rho_{\mathcal{R}}^\star$ yields Equation (19), including its $O(s_t^2)$ remainder.

Finally write $v_t = r_1^2 V_t$ and $\Gamma_t = \rho_{\mathcal{R}}^\star G_t$, where V_t and G_t depend only on the fixed shock and exposure processes. Stationarity and ergodicity give $\mathbb{E}_{\text{time}}[v_t] = r_1^2\mathbb{E}_{\text{time}}[V_t]$ and $\mathrm{SD}_{\text{time}}(v_t) = r_1^2\,\mathrm{SD}_{\text{time}}(V_t)$, with analogous relations involving $\rho_{\mathcal{R}}^\star$ and $|\rho_{\mathcal{R}}^\star|$ for the mean and SD of Γ_t. This proves the proportionalities whenever the corresponding common time moment is nonzero. Replacing $\mathcal{R}$ by $\mathcal{Q}$ and setting $\zeta \equiv 0$ proves the output version. ■

Proof of Corollary 2. Specializing Proposition 6 to $\sigma_\kappa^2 = 0$ gives $\mu_t = -b_R\Delta u_t + O(x^2)$ and, to leading order, $\sqrt{v_t} = |\mathcal{R}'(0)|\sigma_\lambda|\Delta u_t|$. On impact, $\Delta u_1 = x > 0$. During recovery, $\Delta u_t = -(1-\beta)\beta^{t-2}x < 0$ for $t \geq 2$. Mean growth is therefore negative on impact and positive during recovery, while dispersion rises from zero on impact. Their positive leading terms during recovery decay by a factor of β each period.

With $C_\lambda := (3/2)\mathrm{Var}(\lambda^2)/\sigma_\lambda^3 > 0$, the leading skewness term is $\Gamma_t \simeq \rho_{\mathcal{R}}^\star C_\lambda(u_t + u_{t-1})\,\mathrm{sgn}(\Delta u_t)$. On impact it equals $\rho_{\mathcal{R}}^\star C_\lambda x < 0$; for $t \geq 2$ it equals $-\rho_{\mathcal{R}}^\star C_\lambda(1+\beta)\beta^{t-2}x > 0$ and also decays by a factor of β. Finally, holding the shock and exposure processes fixed, $v_t \propto \mathcal{R}'(0)^2$ and $|\Gamma_t| \propto |\rho_{\mathcal{R}}^\star|$. The size comparison follows from the relationships between firm size, policy slope, and normalized curvature established in Section 3.3.3. ■

B.2 Identification Proof

Proof of Proposition 7. The proof proceeds in three stages: use mean growth as a local proxy for the aggregate shock, identify relative slopes and idiosyncratic volatility from the affine variance equation, and remove the resulting group-specific attenuation from skewness before taking RMS ratios.

Average Proposition 6's group mean equation across the ten equal-mass deciles. Demeaning over time removes constants and, under stationarity, the time means of Δu_t, $\Delta(u_t^2)$, and the common residual difference. Since $\Delta(u_t^2) = (u_t + u_{t-1})\Delta u_t$

and both factors are first order in s_t,

$$m_t = -b_m \Delta u_t + \frac{\overline{\mathcal{R}''(0)}}{2}\sigma_\lambda^2(u_t + u_{t-1})\Delta u_t + \Delta \overline{Z}_t + O(s_t^3) = -b_m \Delta u_t + O(s_t^2).$$

Local inversion gives $\Delta u_t = -m_t/b_m + O(s_t^2)$. Squaring yields $(\Delta u_t)^2 = m_t^2/b_m^2 + O(s_t^3)$, while multiplication by $u_t + u_{t-1}$ gives $(u_t + u_{t-1})\Delta u_t = (u_t + u_{t-1})\operatorname{sgn}(\Delta u_t)|m_t|/b_m + O(s_t^3)$. The square has a third-order remainder because it multiplies a first-order leading term by a second-order inversion error.

Within group j, set $r_j := \mathcal{R}_j'(0)$. Proposition 6 gives $v_{jt} = r_j^2[2\sigma_{\kappa,j}^2 + \sigma_\lambda^2(\Delta u_t)^2] + O(s_t^3)$. Define $B_j := r_j^2\sigma_\lambda^2/b_m^2$ and $I_j := 2r_j^2\sigma_{\kappa,j}^2$. Substitution yields $v_{jt} = I_j + B_j m_t^2 + O(s_t^3)$. Taking time-series standard deviations removes I_j, so $\mathrm{SD}_{\mathrm{time}}(v_{jt}) = B_j \mathrm{SD}_{\mathrm{time}}(m_t^2) + O(s^3)$ and $\beta_j = B_j + O(s)$. Consequently, $[\mathrm{SD}_{\mathrm{time}}(v_{jt})/\mathrm{SD}_{\mathrm{time}}(v_{5t})]^{1/2} = (B_j/B_5)^{1/2} + O(s) = |r_j|/|r_5| + O(s)$. Taking time means of the affine representation and subtracting $\beta_j \mathbb{E}_{\mathrm{time}}(m_t^2)$ gives $\alpha_j = I_j + O(s^3)$ and hence $\alpha_j/(2\beta_j) = b_m^2\sigma_{\kappa,j}^2/\sigma_\lambda^2 + O(s^3)$. The ratio with Decile 5 followed by the positive square root proves the relative-volatility formula.

It remains to verify the nonlinear attenuation correction. Proposition 6 gives

$$\Gamma_{jt} = \rho_j \frac{3}{2}\frac{(u_t + u_{t-1})\Delta u_t\left[4\sigma_{\kappa,j}^2\sigma_\lambda^2 + (\Delta u_t)^2 \mathrm{Var}(\lambda^2)\right]}{\left[2\sigma_{\kappa,j}^2 + (\Delta u_t)^2\sigma_\lambda^2\right]^{3/2}} + O(s_t^2).$$

Let $K := \mathrm{Var}(\lambda^2)$ and $\ell := \sigma_\lambda^2$, treating b_m, K, ℓ as positive order-one constants. Then $\Delta u_t = O(s)$, $\sigma_{\kappa,j}^2 = O(s^2)$, $|m_t| = b_m|\Delta u_t| + O(s^2)$, $m_t^2 = b_m^2(\Delta u_t)^2 + O(s^3)$, and $\alpha_j/\beta_j = 2b_m^2\sigma_{\kappa,j}^2/\ell + O(s^3)$.

Write the numerator and denominator of A_{jt} as $N_{jt} := |m_t|[2\alpha_j/\beta_j + (K/\ell^2)m_t^2]$ and $D_{jt} := (K/\ell^2)(\alpha_j/\beta_j + m_t^2)^{3/2}$. The bracket in N_{jt} equals $(b_m^2/\ell^2)[4\sigma_{\kappa,j}^2\ell + K(\Delta u_t)^2] + O(s^3) = O(s^2)$. Multiplication by $|m_t|$ therefore gives

$$N_{jt} = \frac{b_m^3|\Delta u_t|}{\ell^2}\left[4\sigma_{\kappa,j}^2\ell + K(\Delta u_t)^2\right] + O(s^4) =: N_{jt}^0 + O(s^4), \qquad N_{jt}^0 = O(s^3).$$

For the denominator, $X_{jt} := \alpha_j/\beta_j + m_t^2 = (b_m^2/\ell)[2\sigma_{\kappa,j}^2 + \ell(\Delta u_t)^2] + O(s^3) =: X_{jt}^0 + O(s^3)$. Assume the leading variance mixture is nondegenerate, so $X_{jt}^0 = \Theta(s^2)$. Taylor expansion of $x^{3/2}$ around X_{jt}^0 gives $(X_{jt})^{3/2} = (X_{jt}^0)^{3/2} + O(s^4)$, and therefore

$$D_{jt} = \frac{Kb_m^3}{\ell^{7/2}}\left[2\sigma_{\kappa,j}^2 + \ell(\Delta u_t)^2\right]^{3/2} + O(s^4) =: D_{jt}^0 + O(s^4), \qquad D_{jt}^0 = \Theta(s^3).$$

Dividing the two expansions yields

$$A_{jt} = \frac{|\Delta u_t|\ell^{3/2}}{K}\frac{4\sigma_{\kappa,j}^2\ell + K(\Delta u_t)^2}{\left[2\sigma_{\kappa,j}^2 + \ell(\Delta u_t)^2\right]^{3/2}} + O(s).$$

The fourth-order input errors become first order because the denominator is order s^3. Set $C_\lambda := (3/2)K/\ell^{3/2}$. Since $u_t + u_{t-1} = O(s)$, the first-order error in A_{jt} becomes

second order after multiplication by $u_t + u_{t-1}$. Comparing the preceding expression with the skewness equation gives $\Gamma_{jt} = \rho_j C_\lambda (u_t + u_{t-1}) \operatorname{sgn}(\Delta u_t) A_{jt} + O(s_t^2)$. If A_{jt} is bounded away from zero, division is regular and $\Gamma_{jt}/A_{jt} = \rho_j C_\lambda (u_t + u_{t-1}) \operatorname{sgn}(\Delta u_t) + O(s_t^2)$. On common dates, the time series multiplying ρ_j is identical across groups. The reverse triangle inequality for the RMS norm, $|\operatorname{RMS}(X + R) - \operatorname{RMS}(X)| \leq \operatorname{RMS}(R)$, preserves the uniform second-order remainder. Therefore

$$\operatorname{RMS}_{\text{time}}(\Gamma_{jt}/A_{jt}) = |\rho_j| C_\lambda \operatorname{RMS}_{\text{time}}[(u_t + u_{t-1}) \operatorname{sgn}(\Delta u_t)] + O(s^2).$$

If the common RMS factor is $\Theta(s)$ and $\rho_5 \neq 0$, taking the ratio with Decile 5 proves the relative-curvature formula with remainder $O(s)$. ■

C Data Preparation

For both Compustat samples, calendar quarters come from `datacqtr`. We require a valid `datadate` and retain the latest record within duplicate firm-quarter pairs. Growth is the log change between positive reported sales observations exactly four quarters apart.

C.1 Compustat Quarterly Sample

Sample restrictions. We start from quarterly Compustat North America and retain industrial-format, standard, domestic, consolidated records (`indfmt`=`INDL`, `datafmt`=`STD`, `popsrc`=`D`, and `consol`=`C`). Firms must be incorporated in the United States (`fic`=`USA`), observations must be reported in U.S. dollars (`curcdq`=`USD`), the current company-header SIC must be nonmissing and below 9000, and NAICS must be available. The baseline quarterly sample covers 1984Q1–2024Q4.

Real sales and growth rates. Nominal quarterly sales are deflated by the GDP price deflator (FRED series `GDPDEF`) to obtain real sales.

Cleaning and outlier treatment. The cleaned growth sample requires positive current-quarter sales and assets, usable year-on-year log growth, and the four-quarter sales history needed to measure firm size. We trim the top and bottom 1 percent of log growth observations each quarter, then retain observations with a valid within-industry size classification in two-digit NAICS industries with at least ten firms that quarter.

Table C.1: Summary Statistics for Compustat Data

	Full Compustat	Cleaned Sample	QFR
Assets (mln. USD)	4,170	4,267	43.2
Sales (mln. USD)	414.2	423.3	10.8
Sales Growth (%)	7.3	7.2	0.63
25th percentile of Sales Growth (%)	-7.8	-7.1	-25.3
75th percentile of Sales Growth (%)	20.8	20.1	26.6
Net Leverage (%)	23.4	19.1	20.0
Short-term debt (%)	33.9	33.3	33.0
Obs./quarter	6,392	6,143	6,122
Unique firms	21,089	20,355	–
ρ(Sales Gr., GDP Gr.)	0.75	0.76	–
ρ(Sales Gr., Skew)	0.36	0.77	–
ρ(Skew, GDP Gr.)	0.18	0.51	–

Note: QFR: manufacturing, 1977Q3–2014Q1; statistics from Crouzet and Mehrotra (2020), Tables 1 and 3. Compustat: 1984Q1–2014Q1; sample cleaning follows Appendix C.1. Assets and sales are in 2009 dollars; Skew denotes cumulant skewness.

Firm size. We measure firm size in quarter t by average real sales over the preceding four quarters, requiring positive reported sales in each. Each quarter, we sort firms in the cleaned growth sample into size deciles within two-digit NAICS industries. For robustness, we also rank the same firms economy-wide. Subsample analyses retain these rankings unless stated otherwise.

Sample diagnostics. The four-quarter sales-history requirement excludes a further 1.8% of otherwise usable growth observations before trimming. The sample mainly covers firms with established reporting histories: the median observation is 11.0 years after a firm first appears in Compustat, and only 3.8% fall within its first two years.

On average, firms appearing for the first or last time account for 1.8% and 2.0% of quarterly observations, respectively. These shares measure sample turnover rather than economic entry and exit, since firms may leave the sample when they no longer meet the inclusion criteria.

Sample comparison. Table C.1 compares the raw and cleaned Compustat growth samples with the Quarterly Financial Reports sample used by Crouzet and Mehrotra (2020).

C.2 Compustat Global Quarterly Sample

Universe and reporting conventions. We complement the North America sample with quarterly accounting data from Compustat Global over 2000Q1–2024Q4. We retain firms incorporated outside the United States, industrial-format consolidated statements reported under Compustat's historical standard, and observations with valid SIC codes below 9000 and two-digit NAICS codes.

Sales growth. We deflate quarterly sales using country-specific annual GDP deflators. Country and currency reporting must be consistent across the two quarters used to measure growth. We trim the top and bottom 1 percent of growth observations within each country-quarter.

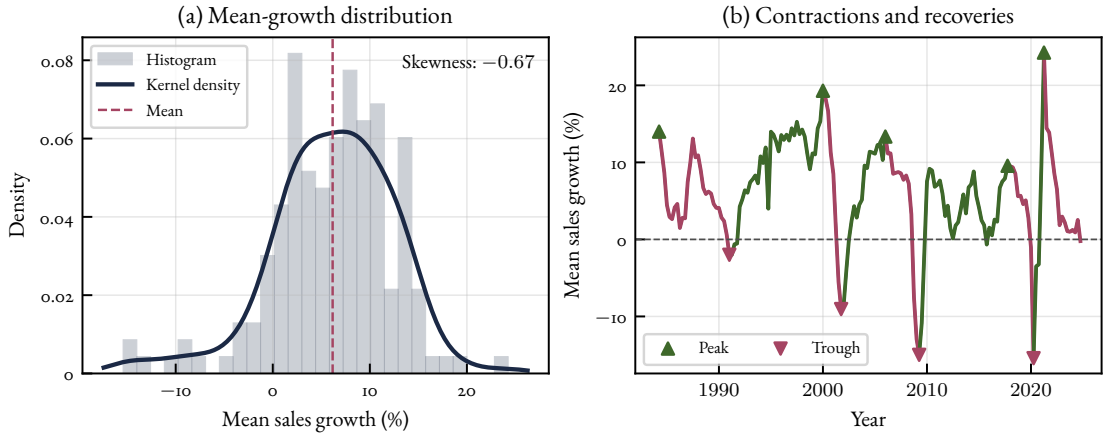
Accounting filters and size. As in the U.S. sample, we require positive sales and assets and measure firm size by average real sales over the preceding four quarters. Sales must be positive throughout this history, with the same country and currency as the current observation. Each quarter, we assign firms to size deciles within two-digit NAICS industries in each country, requiring at least ten firms per group. Unlike in the U.S. sample, these rankings are formed before restricting observations to usable growth rates and trimming growth tails. We retain only observations with a valid size classification.

D Additional Empirical Results

D.1 Aggregate Sales-Growth Turning Points

Panel (a) of Figure D.1 shows the negatively skewed distribution of mean sales growth, with cumulant skewness of -0.67. Panel (b) traces the sharp contractions and subsequent recoveries, including the large positive growth spikes following the 2009 and 2020 troughs. We identify turning points using an adaptation of the algorithm in Dupraz et al. (2025). The algorithm tracks a running maximum until growth falls by at least fourteen percentage points, confirming a peak. It then tracks a running minimum until growth rises by the same amount, confirming a trough.

Figure D.1: Mean Growth Distribution and Turning Points



Note: Quarterly mean real-sales growth, 1984Q1–2024Q4. Panel (a) shows its distribution and kernel density; the dashed line marks its time average. Panel (b) shows declines in red and recoveries in green, with markers at confirmed peaks and troughs. Growth is in log-growth percentage points.

D.2 Economy-Wide Size Gradients

Figure D.2 repeats Figure 3 using economy-wide size deciles. The sample and size measure follow Appendix C.1 and all other moment and inference choices are identical to Figure 3.

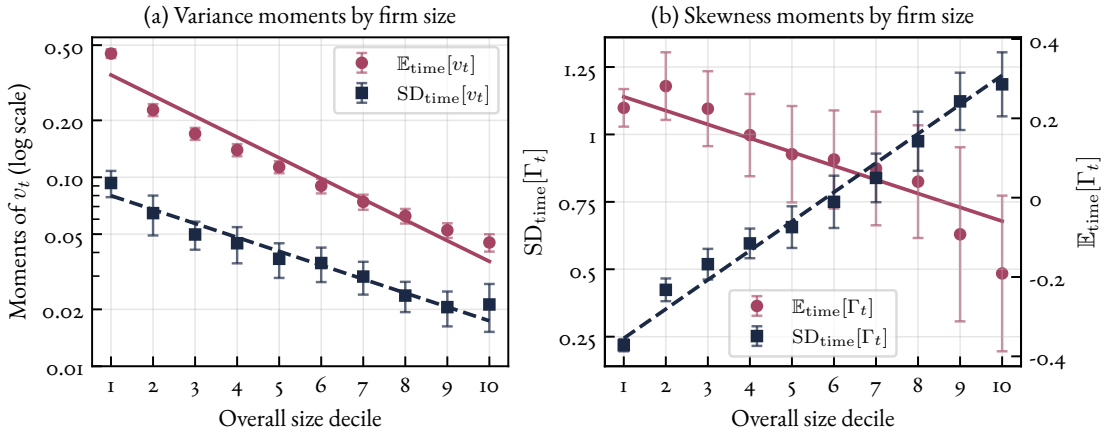
Both the time-series mean and standard deviation of growth variance fall with economy-wide size, whereas the time-series standard deviation of cumulant skewness rises sharply.

D.3 Compustat Global Results

We assess the external validity of the main stylized facts using the non-U.S. Compustat Global sample described in Appendix C.2, covering 2000Q1–2024Q4. For both exercises below, we calculate moments within countries and then average them with equal country weights. We require at least 50 firms per country-quarter and at least 20 qualifying quarters per country, yielding 40 countries and 2,854 country-quarter observations.

Figure D.3 repeats the analysis in Figure 2, expressing each country's mean growth, variance, and skewness as deviations from its own time-series average. The raw time-series correlation between the quarterly averages of centered growth and skewness is 0.56. Panel (b) groups the country-quarter observations into 50 equal-count bins and plots the mean growth and variance in each bin. As in Figure 2, we impose a symmetric V-shaped fit by regressing country-adjusted variance

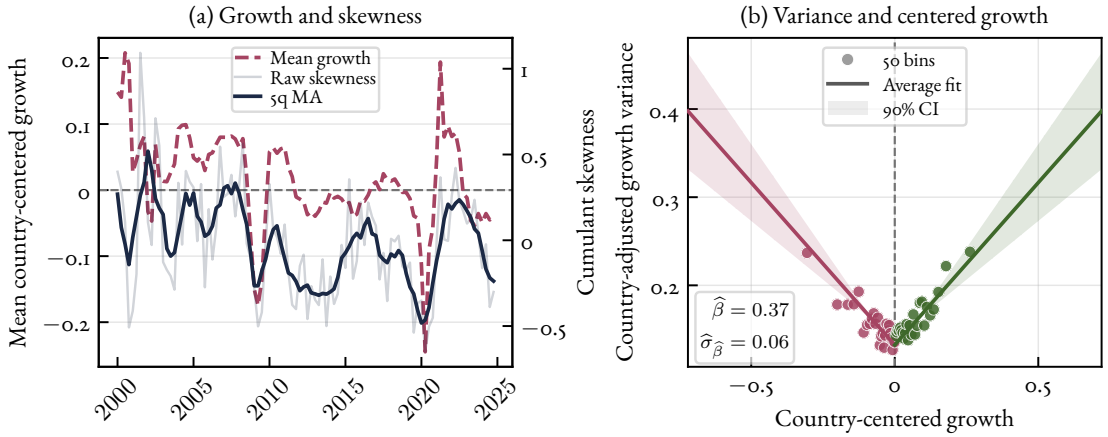
Figure D.2: Economy-Wide Opposing Size Gradients in Growth Moments



Note: Economy-wide size deciles, 1984Q1–2024Q4. See the note to Figure 3 for symbols, confidence intervals, and fitted lines.

on an intercept and absolute country-centered growth. The regression uses the underlying country-quarter observations and yields $\widehat{\beta} = 0.368$. Confidence bands use standard errors clustered by country.

Figure D.3: Growth, Skewness, and Dispersion in Compustat Global

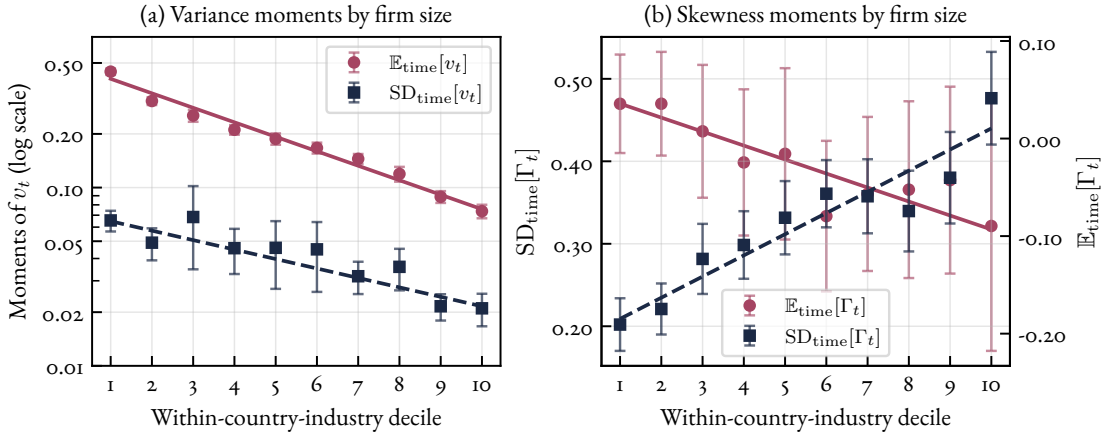


Note: Compustat Global, 2000Q1–2024Q4. Panel (a) equally weights country-adjusted moments: dashed red is mean growth; solid blue is a centered five-quarter average of skewness. Panel (b) shows 50 equal-count bins and a symmetric V-shaped fit with 90% country-clustered confidence bands. Growth is in log units; variance is in squared log units.

Figure D.4 reports the size gradients for this sample. Size deciles follow Appendix C.2. We use unstandardized real-sales growth and calculate moments within each country after pooling firms with the same size decile. Moments require at least 30 firms per country-quarter-decile, and quarterly averages require at least five countries. These requirements yield 21 contributing countries over 2005Q1–2024Q4.

Panel (a) shows that both the time-series mean and time-series standard deviation of growth variance broadly decline with firm size. Panel (b) shows a noisy decline in mean skewness and a broad increase in its time-series standard deviation across size deciles.

Figure D.4: Growth Variance and Skewness by Firm Size in Compustat Global

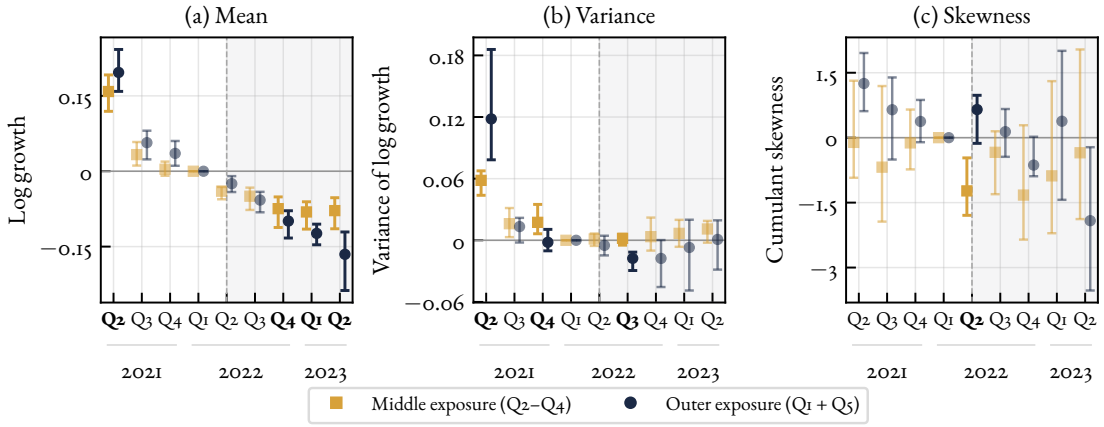


Note: Countries receive equal weight within each quarter-decile. See the note to Figure 3 for symbols, confidence intervals, and fitted lines.

D.4 Inflation Exposure and Growth-Distribution Changes

Figure D.5 applies the procedure in Section 5.4 to the inflation exposures of Davis et al. (2025), comparing outer exposure quintiles (1 and 5) with middle quintiles (2–4). The balanced sample spans 2021Q2–2023Q2 and contains 1,086 firms across nine industries. We measure size and moment changes relative to 2022Q1, holding exposure groups, industry–size cells, and common composition weights fixed.

Figure D.5: Inflation Exposure and Growth-Distribution Changes



Note: Changes from 2022Q1; shading begins at 2022Q2. Mean and variance use log and squared log units. See the note to Figure 7 for symbols, exposure groups, weighting, and inference.

In 2022Q2, variance changes little in either group, while skewness rises among outer-exposure firms and falls among middle-exposure firms. The outer-minus-middle variance contrast is -0.006, with a 90 percent pointwise interval of $[-0.021, 0.009]$. The skewness contrast is positive at 1.88, with an interval of $[0.67, 2.52]$, reversing the COVID ordering.

Two features limit what we can learn from this reversal. First, the variance paths already differ before 2022Q2. Pre-event skewness contrasts, by comparison, do not individually exclude zero at the 10 percent level. Second, inflation rose over several quarters, overlapping the pandemic recovery and multiple news events. The exposure measure itself combines equity-market reactions in May and September 2022, including news about earnings and expected interest rates as well as inflation.

D.5 Risk-Exposure Moments by Firm Size

We examine the exposure moments discussed in Section 4.3. We use the DHS pandemic and inflation exposures of Davis et al. (2025), centered within each shock and size group, as proxies for the model's exposure loadings. Their return-based sign convention need not coincide with that of the model's cost loadings.

We compare exposure variance and $\mathbb{V}[\lambda^2]/\mathbb{V}[\lambda]^{3/2}$ across size deciles. When idiosyncratic risk is negligible, the latter equals $(2/3)C_\lambda$ in Section 4.3. We also report exposure skewness to assess the symmetry assumption in Proposition 6.

We match the untrimmed DHS exposures to the Compustat sample and use the within-industry size percentiles defined in Appendix C.1. The reference quarters are 2020Q1 for COVID and 2022Q2 for inflation, so the underlying sales history

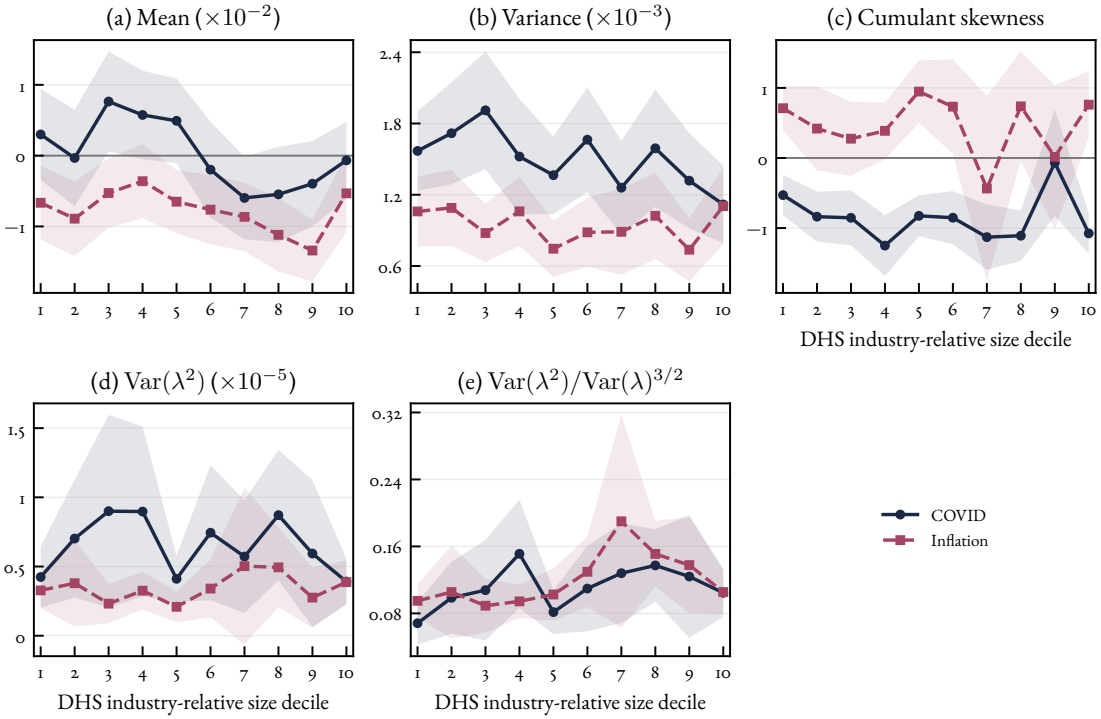
precedes the respective events. Separately for each measure, we sort matched firms by their size percentile into ten equal-count groups. The samples contain 1,432 COVID and 1,447 inflation firms.

Let x_{ic} denote the raw DHS proxy and $\bar{x}_{cj}$ its mean within shock c and size decile j. For firms in that group, define the centered proxy $\lambda_{ic} = x_{ic} - \bar{x}_{cj}$. Panel (a) of Figure D.6 reports the raw-proxy mean $\bar{x}_{cj}$; the remaining panels report variance, cumulant skewness, $\mathbb{V}[\lambda^2]$, and $\mathbb{V}[\lambda^2]/\mathbb{V}[\lambda]^{3/2}$. Bootstrap inference resamples firms within fixed shock–size groups and recenters exposures in each draw.

The exposure moments are non-monotonic across size deciles, and their confidence intervals are consistent with equal values across deciles. The evidence therefore does not establish the size ordering required by an exposure-based explanation.

COVID exposures are left-skewed in every bin, whereas inflation exposures are right-skewed in nine of the ten groups. Exposure skewness has no systematic ordering by firm size.

Figure D.6: DHS exposure moments by firm size



Note: Ten equal-count groups per exposure measure, ordered by within-industry size percentile. Panel (a) reports raw-proxy means; the remaining panels use exposures centered within each shock and size group. Shaded regions are pointwise 95% normal intervals using standard errors from 1,999 firm-bootstrap draws within fixed groups.

E Responses to Aggregate Shocks: Estimation and Inference

This appendix gives the estimation and inference details for Section 5.3.

E.1 Shock Sources and Controls

Selected innovations. The main business-cycle shock is the original replication-package construction of Angeletos et al. (2020), ending in 2017Q4. Its sign is positive when unemployment rises. Credit innovations follow the excess-bond-premium construction of Gilchrist and Zakrajšek (2012): a recursive quarterly VAR with two lags, ordered as consumption growth, nonresidential investment growth, GDP growth, GDP-deflator inflation, the excess bond premium, stock-price growth, the ten-year Treasury yield and the federal funds rate. We refit the VAR through 2019Q4 and use its orthogonalized excess-bond-premium innovation.

Oil supply news uses the pre-Covid monthly series of Känzig (2021), summed over three complete months. Monetary policy uses the U.S. monetary component of Jarociński and Karadi (2020), which separates policy tightening from central-bank information using interest-rate and stock-price surprises. We use the updated author decomposition restricted to 1994–2019. Investment news follows Ben Zeev and Khan (2015): we refit the nine-state VAR through 2019Q4, retaining its four lags and 60-quarter news-identification objective. Financial uncertainty uses the max-G financial shock of Ludvigson et al. (2021), refitted through December 2019 with CPI-deflated gold returns in the identifying input. The refits retain the authors' identifying restrictions and use revised historical inputs.

Signs, scales and dates. For innovation ϵ_t^s, set

$$x_t^s = \omega_s \epsilon_t^s / \widehat{\mathrm{SD}}(\epsilon^s), \qquad \omega_s \in \{-1, 1\}, \tag{E.1}$$

with ω_s chosen so positive values denote adverse innovations. The scale is the population standard deviation over 1984Q1–2019Q4, except 1984Q1–2017Q4 for the main business cycle and 1994Q1–2019Q4 for monetary policy. It is fixed across firms, horizons and bootstrap draws. The adverse component is $\max\{x_t^s, 0\}$; the favorable component is $\min\{x_t^s, 0\}$, so the response to a negative one-unit innovation is minus its coefficient. Both components enter jointly.

Macro controls. Every firm LP contains two lags of each state and two lags of own growth, with spell-specific slopes. All shocks use log real GDP, log GDP deflator and the excess bond premium in levels.

E.2 Firm Spells and Relative Responses

Estimation sample. A *spell* is an uninterrupted sequence of quarterly observations for one firm. We apply the accounting restrictions and quarterly growth trim in Appendix C. Growth, sales-size histories, industry codes, shocks and controls must be observed at the dates needed for the regression. A missing or growth-trimmed observation ends a spell; subsequent usable observations form a new spell. A change in the firm's size group does not end a spell.

Within each shock's sample window, we retain every such sequence containing at least 40 quarters of usable growth. A firm can contribute several nonoverlapping spells, each of which is estimated separately using the full uninterrupted sequence. A length-L spell supplies $L-2-h$ observations at horizon h. Lags and forward outcomes are constructed within each spell.

The firm-specific design must have full column rank and positive residual degrees of freedom at $h=6$; earlier-horizon designs contain those rows. The roster uses the common support of the light and paper-control specifications considered during sample construction. The same spells enter every horizon, while observation counts decline with h.

Size groups for the moment comparison. We use quarterly within-two-digit-NAICS ranks of average real sales over $t-4,\ldots,t-1$, computed in the broad accounting sample. To enter the small-firm (large-firm) group, the firm must have an observed rank in every quarter of the spell and remain in the lower (upper) half throughout. This includes quarters used for lags or forward outcomes. Spells in which a firm crosses the median or lacks ranks in some quarters still enter response estimation and the average response used to select the horizon of the largest contraction. They are excluded from the comparison of response moments across size groups. A firm can contribute an earlier spell to the small-firm group and a later spell to the large-firm group. Industry membership follows the firm's reported code at each date.

Time fixed effects and normalization. Equation (24) is joint OLS with spell-specific intercepts, slopes, and common time fixed effects. Because the shocks vary

Table E.1: Firm-Spells and Stable Size Groups

	Firms	Spells	Stable small	Stable large	Reporting spells
Main business cycle	6,720	6,936	2,011	2,505	4,470
Credit spread	6,860	7,102	2,065	2,541	4,555
Oil supply news	6,860	7,102	2,065	2,541	4,534
Monetary policy	5,125	5,181	1,557	1,963	3,488
Investment news	6,860	7,102	2,065	2,541	4,535
Financial uncertainty	6,860	7,102	2,065	2,541	4,535

Note: Firms and spells count all estimated responses. Stable-size columns count spells before cell eligibility. Reporting spells count distinct spells in the selected-horizon reporting cells. A firm may contribute more than one spell.

only over time, a common shift in the signed shock slopes can be absorbed by the time fixed effects. We normalize the equal-spell mean of each signed slope to zero. This identifies relative responses and leaves cell-centered variance and skewness unchanged.

Selecting the horizon of the largest average contraction. To estimate the average response needed for horizon selection, we use an auxiliary regression that omits time fixed effects but otherwise uses the same observations and regressors. The response moments are computed from the main regression with time fixed effects. With J_s estimated spells, choose

$$h_s^\star = \arg\min_{h\in\{0,\dots,6\}} \frac{1}{J_s}\sum_{j=1}^{J_s} \widehat{b}_{jh}^{s,+,\text{aux}}, \tag{E.2}$$

breaking ties at the earliest horizon. Every estimated spell has equal weight, including movers and spells with incomplete size labels. We select the horizon using untrimmed mean responses before forming the size-group moment contrasts. Selected horizons are 1, 3, 5, 4, 5 and 3 for the main business-cycle, credit, oil, monetary-policy, investment-news and uncertainty shocks, respectively.

E.3 Cell Moments and Aggregation

We retain all estimated coefficients without coefficient trimming; the quarterly growth trim used to construct the regression sample is unchanged.

Reporting support. We form industry–date cells at adverse-shock dates, using observations available at the selected horizon $h_s^\star$. We require ten stable-small and ten stable-large firms per cell and at least five qualifying industries per date. We

hold these cells and their aggregation weights fixed across bootstrap draws. All point and bootstrap cells have positive raw variance. A spell coefficient can appear at several dates, so its total weight reflects its presence in the eligible reporting cells.

Variance adjustment. Estimated shock responses contain sampling error. Even if true responses were identical within a cell, their estimates would be dispersed. This source of variation matters for size comparisons when response estimates differ in precision across small and large firms.

Suppress shock and horizon indices. For a fixed cell I of n firms, write $\widehat{\boldsymbol{b}}_I = \boldsymbol{b}_I + \boldsymbol{e}_I$, where $\boldsymbol{b}_I$ contains true adverse-shock responses and $\boldsymbol{e}_I$ their estimation errors. Let $M_I = I_n - \mathbf{1}_n\mathbf{1}_n'/n$. If coefficient errors are centered conditional on the true responses and have covariance $\Omega_{e,I}$, then

$$\begin{aligned}\mathbb{E}\left[\frac{1}{n}\widehat{\boldsymbol{b}}_I' M_I \widehat{\boldsymbol{b}}_I \,\middle|\, \boldsymbol{b}_I\right] &= \frac{1}{n}\boldsymbol{b}_I' M_I \boldsymbol{b}_I + \frac{1}{n}\mathbb{E}[\boldsymbol{e}_I' M_I \boldsymbol{e}_I \mid \boldsymbol{b}_I] \\ &= \frac{1}{n}\boldsymbol{b}_I' M_I \boldsymbol{b}_I + \frac{1}{n}\operatorname{tr}(M_I \Omega_{e,I}). \end{aligned} \tag{E.3}$$

The cross term vanishes because coefficient errors are centered. The trace term is the contribution of coefficient estimation error to centered coefficient variance, including covariance between firms' errors. Replacing $\Omega_{e,I}$ with its estimate gives the adjusted variance in Equation (25):

$$\widehat{V}_I^{\mathrm{adj}} = \frac{1}{n}\widehat{\boldsymbol{b}}_I' M_I \widehat{\boldsymbol{b}}_I - \frac{1}{n}\operatorname{tr}(M_I \widehat{\Omega}_{e,I}). \tag{E.4}$$

The first term, denoted $V_I(\widehat{\boldsymbol{b}})$, is the observed dispersion of estimated responses. Appendix E.4 describes how the calendar-block procedure supplies the covariance estimate and lower confidence bounds. All moments use denominator n.

If the estimated contribution of coefficient error exceeds observed variance, the adjusted variance is negative. We retain such estimates.

Size-group contrasts. For moment $M \in \{V^{\mathrm{adj}}, \Gamma\}$, let T_s be the retained adverse dates and K_{st} their eligible industries. Report

$$\Delta M_s^\star = \frac{1}{|T_s|}\sum_{t\in T_s}\frac{1}{|K_{st}|}\sum_{k\in K_{st}}\left(\widehat{M}_{skt,\mathrm{small}} - \widehat{M}_{skt,\mathrm{large}}\right). \tag{E.5}$$

Skewness is calculated within each cell before averaging the small-minus-large differences.

E.4 Bootstrap Inference and Coefficient Reliability

Calendar-block draws. We use a blockwise wild construction (Yeh, 1998; Shao, 2010) with OLS residuals. Divide the master calendar into nonoverlapping six-quarter blocks beginning in 1984Q1. For each of 1,000 draws, assign independent Rademacher signs to the 24 blocks through 2019Q4. Signs are shared across firms, spells, horizons and shocks, preserving common movements across firms and dependence within calendar blocks.

For each draw, multiply the OLS residuals by the assigned block signs and add them to the fitted outcomes. Refit Equation (24) on these outcomes, re-estimating the shock coefficients, nuisance coefficients and time fixed effects with the original regressors and normalizations. Let $\widehat{\boldsymbol{b}}^{*(r)}$ contain the resulting adverse-shock coefficients and define their deviations from the original estimates as $\boldsymbol{e}^{(r)} = \widehat{\boldsymbol{b}}^{*(r)} - \widehat{\boldsymbol{b}}$. We estimate $\widehat{\Omega}_{e,I}$ from the covariance of these bootstrap deviations within each retained cell I.

For the fixed set I of coefficients, with $n = |I|$, the variance and skewness draws are

$$\begin{aligned} V_I^{*(r)} &= V_I(\widehat{\boldsymbol{b}} + \boldsymbol{e}^{(r)}) - V_I(\boldsymbol{e}^{(r)}) - \frac{1}{n}\operatorname{tr}(M_I \widehat{\Omega}_{e,I}), \\ \Gamma_I^{*(r)} &= \Gamma_I(\widehat{\boldsymbol{b}} + \boldsymbol{e}^{(r)}). \end{aligned} \tag{E.6}$$

Perturbing the coefficients adds variance. The variance draw subtracts the realized variance of that perturbation and the original plug-in variance correction, computing both on the fixed cell. We aggregate the small-minus-large contrasts with the original cell weights. Standard errors are the sample standard deviations of the 1,000 contrast draws. For $H_0 : \Delta \leq 0$ against $H_1 : \Delta > 0$, the one-sided confidence-c lower bound is the point estimate minus z_c standard errors, where z_c is the standard-normal quantile and $c \in \{0.90, 0.95, 0.99\}$. These pointwise bounds condition on the source shocks, regressors, firm-spell sample, size classifications and selected horizon.

Skewness and coefficient reliability. We report the unadjusted skewness of the retained coefficients,

$$\widehat{\Gamma}_I = \frac{n^{-1}\sum_{j \in I}(\widehat{b}_j - \bar{\widehat{b}}_I)^3}{V_I(\widehat{\boldsymbol{b}})^{3/2}}, \tag{E.7}$$

where $\bar{\widehat{b}}_I$ is the cell mean. For a randomly drawn firm j within a size–industry–date group, write $\widehat{b}_j = b_j + e_j$, where b_j is the true response and e_j its estimation

error. Adding independent, centered, unskewed estimation error leaves the third central moment of the response unchanged but increases its variance from $V(b_j)$ to $V(b_j) + V(e_j)$. Thus

$$\Gamma(\widehat{b}_j) = r^{3/2}\Gamma(b_j), \qquad r := \frac{V(b_j)}{V(b_j) + V(e_j)} = \frac{\mathrm{SNR}}{1 + \mathrm{SNR}}, \qquad \mathrm{SNR} := \frac{V(b_j)}{V(e_j)}. \quad \text{(E.8)}$$

Equal reliability across the two size groups within a cell multiplies their skewness difference by the same positive factor, preserving its sign.

For a fixed cell with positive estimated signal, estimated reliability is $\widehat{r}_I = \widehat{V}_I^{\mathrm{adj}}/V_I(\widehat{\boldsymbol{b}})$: the share of observed coefficient variance attributed to response heterogeneity. Table E.2 reports the shares of variance attributed to estimation error and the attenuation factors, both formed from the weighted group variance totals.

Table E.2: Estimation Error and Skewness Attenuation

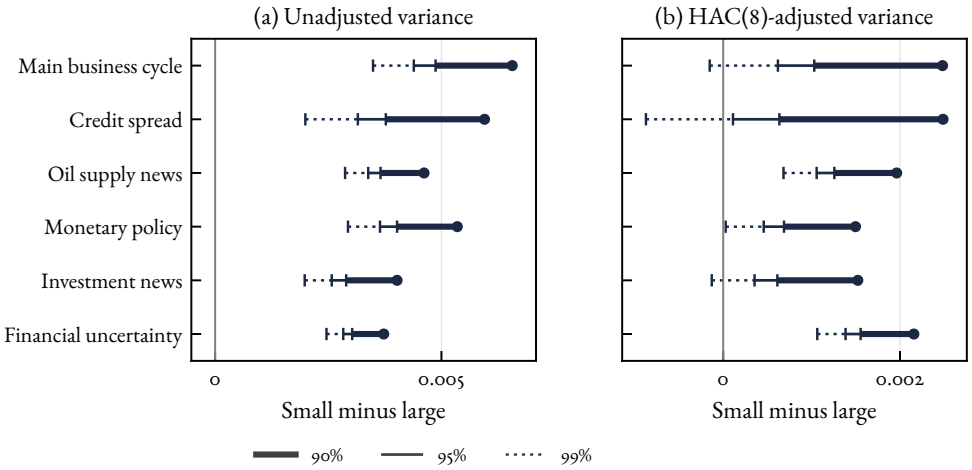
	Error share (percent)		Attenuation factor	
Shock	Small	Large	Small	Large
Main business cycle	68.6	76.0	0.176	0.118
Credit spread	60.8	53.4	0.245	0.318
Oil supply news	64.3	69.0	0.213	0.173
Monetary policy	78.2	72.8	0.102	0.142
Investment news	64.3	59.0	0.213	0.262
Financial uncertainty	44.3	46.7	0.415	0.389
Average			0.227	0.234

Note: Diagnostics use the baseline six-quarter calendar-block variance correction and weighted group variance totals. The error share is the fraction of observed variance attributed to coefficient estimation error. Attenuation is the corrected-to-observed variance ratio to the power 3/2. The latter is a benchmark under independent, centered, unskewed coefficient error, not an exact correction to average cell-level skewness. The final row averages attenuation factors equally across the six shocks.

E.5 Robustness Comparisons

We compare alternative variance corrections using the baseline estimates, reporting cells, weights, and bootstrap procedure. Figure E.1 reports unadjusted variance on the left and HAC(8)-adjusted variance on the right. The unadjusted bootstrap variance is $V_I(\widehat{\boldsymbol{b}} + \boldsymbol{e}^{(r)})$. HAC(8) replaces the plug-in covariance of coefficient errors with the Bartlett-weighted covariance of calendar-quarter coefficient scores through eight lags. All six variance contrasts have positive 95 percent lower bounds under both alternatives.

Figure E.1: Unadjusted and HAC(8)-Adjusted Coefficient Variance



Note: Small-minus-large variance contrasts using the same coefficients and groups as Figure 6. Variance is unadjusted (left) or HAC(8)-adjusted (right). See that figure's note for aggregation and inference. Variance is in squared log-growth units; panel axes use different scales.

F Relative Policy Estimates

Estimation. We apply Proposition 7 to quarterly growth moments in the baseline within-industry size deciles, requiring at least 30 firms per cell.[34] Mean growth m_t is demeaned over the full sample before selecting estimation dates. We retain the 90% of balanced quarters with the smallest m_t^2, leaving 147 quarters.

To limit noise in the curvature adjustment, we fit a smooth positive curve to the estimated variance ratios:

$$\left(\frac{\widehat{\alpha}_j}{2\widehat{\beta}_j}\right)^F = \exp(a + bj + cj^2 + dj^3). \tag{F.1}$$

The fit uses nonlinear least squares in levels. Fitted values enter only the adjustment factor A_{jt}; relative idiosyncratic volatility uses the raw ratios. Curvature comparisons require $A_{jt} \geq 0.05$ in both the relevant decile and Decile 5, with at least eight qualifying quarters. Each baseline curvature comparison retains 139 of the 147 quarters.

The upward curvature gradient persists when we vary this cutoff between 0.01 and 0.075, although individual estimates change. Varying exposure shape over $\mathcal{K}_\lambda \in \{1, 2, 4\}$ has little effect on the estimates. The cutoff comparisons reselect dates; the exposure-shape comparisons keep the baseline dates.

[34] We use sample variance and the standardized third central moment.

Table F.1: Rank Statistics for Relative Policy Estimates

	Spearman			Kendall		
Outcome	ρ_s	90% range	Opp. sign	τ_b	90% range	Opp. sign
Policy slope	-0.988	$[-1.000, -0.952]$	< 0.001	-0.956	$[-1.000, -0.867]$	< 0.001
Idiosyncratic volatility	-0.952	$[-0.976, -0.758]$	< 0.001	-0.867	$[-0.911, -0.556]$	< 0.001
Curvature magnitude	0.939	$[0.709, 0.976]$	< 0.001	0.822	$[0.556, 0.911]$	< 0.001

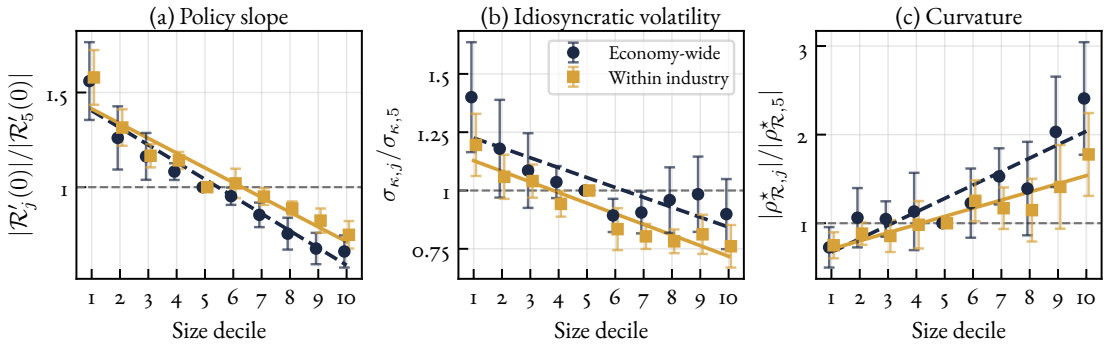
Note: Central 90% ranges from 50,000 joint log-Gaussian draws using Newey–West covariance estimates with four calendar-quarter lags. Opp. sign denotes the share of draws with the opposite sign or zero. Estimates are normalized at Decile 5.

Confidence Intervals. We use the delta method with Newey and West (1987) covariance estimates and four calendar-quarter lags. We calculate covariances from the moments' influence functions, using numerical derivatives for the fitted curve. This accounts for dependence between each estimate and its Decile-5 reference and, for curvature, uncertainty in the fitted variance ratios. The 90% normal intervals condition on the selected dates and full-sample centering of mean growth.

Rank Statistics. We use the estimated covariance matrix of the log estimates to draw 50,000 times from a joint log-Gaussian approximation (Krinsky and Robb, 1986). In each draw, we recompute Spearman's correlation and Kendall's τ_b. Drawing in logs preserves positivity, and Decile 5 remains fixed at one.

Industry Composition. The same three estimated patterns also appear when we rank firms across the whole economy (Figure F.1).

Figure F.1: Economy-Wide and Within-Industry Relative Policy Estimates



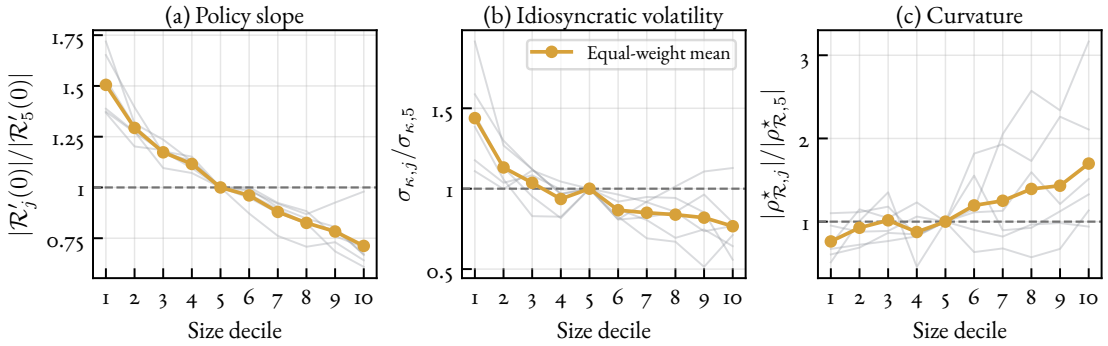
Note: Blue circles use economy-wide size ranks; yellow squares use ranks within two-digit NAICS industries. See the note to Figure 5 for normalization, confidence intervals, and fitted lines.

Larger industries can still dominate the pooled moments. We therefore estimate

the system separately in broad industry groups, preserving the within-NAICS2 size ranks, and give each group's estimates equal weight. Each group is normalized at its own Decile 5 before averaging. Six groups have enough observations for this comparison.[35]

Figure F.2 shows that all three estimated patterns also appear in the equal-weight averages. The same directions persist with a 30-firm cell minimum or estimation by individual two-digit industry.

Figure F.2: Relative Policy Estimates by Industry



Note: Gray lines show estimates for six broad industry groups; the yellow line is their equal-weight mean. Size ranks are within two-digit NAICS industries, with each group normalized at Decile 5.

[35]We require 20 firms per cell, 40 balanced quarters below the common 90th-percentile cutoff of economy-wide m_t^2, and finite estimates for all ten deciles. The retained groups cover natural resources and mining; manufacturing; trade, transportation, and utilities; information; finance and real estate; and professional and business services.